Nitrogen Disposal and The Urea Cycle I & II

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Fall 2026

Biochemistry

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Session Objectives -part 1

1. Identify transport systems for amino acids across cell membranes and explain the consequences of known AA transporter deficiencies.
2. Describe the two steps of nitrogen removal from amino acids prior to the metabolism of their carbon skeletons.
3. Elaborate on the application of serum aminotransferase levels in clinical diagnostics.
4. Explain how ammoniais transported from peripheral tissues to the liver and outline the role of liver (and kidney) in its disposal.

Session Objectives - part 2

1. Outline the sequence of reactions in the urea cycle and trace the flow of nitrogen from amino acids into and out of the cycle.

2. Identify the rate-limiting regulatory step in the urea cycle and explain its regulation.

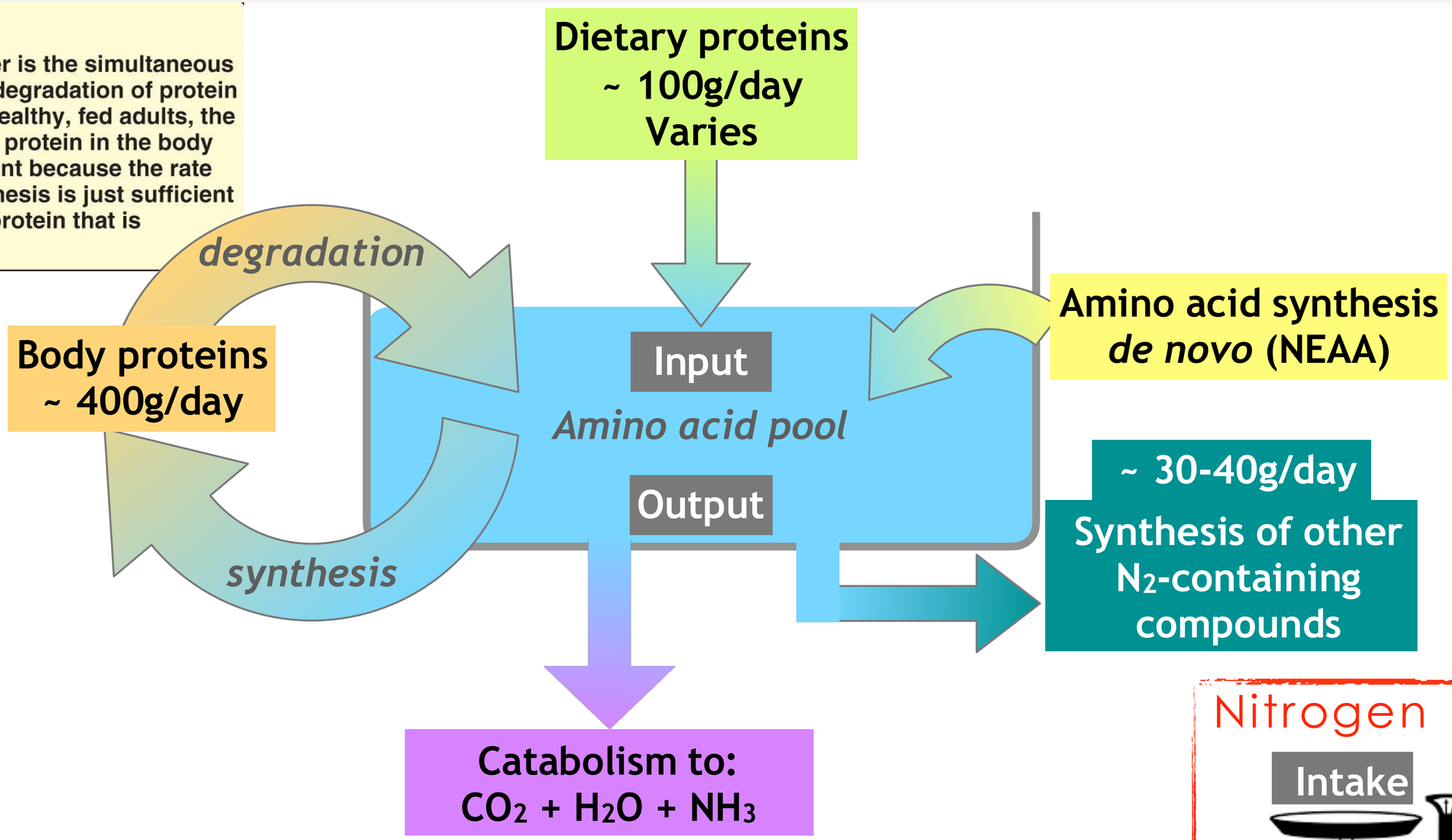
3. Explain under what circumstances hyperammonemia may develop, what are the symptoms and how it could be treated.

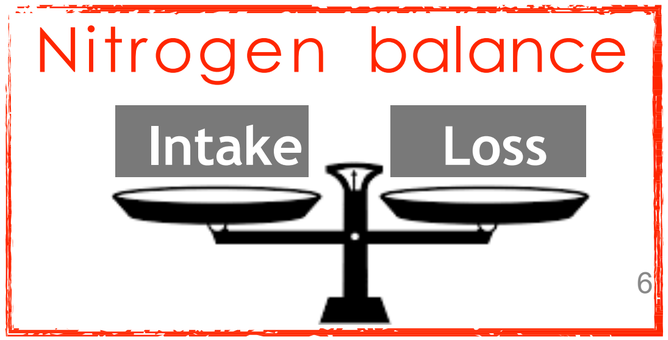
4. Describe the role of kidney in nitrogen disposal.

The Amino Acid Pool and N_2 Balance (Review)

TURNOVER

Protein turnover is the simultaneous synthesis and degradation of protein molecules. In healthy, fed adults, the total amount of protein in the body remains constant because the rate of protein synthesis is just sufficient to replace the protein that is degraded.





Amino Acid Transport (Review)

Transport of AA into the cells:

- **Primarily secondary active transport, often NA^+-dependent (indirectly ATP-dependent).**
- **Some facilitated diffusion also occurs**
- **Multiple transport systems have overlapping AA specificities.**
- Small intestine and the proximal tubule of the kidney **have common transport systems** for amino acid uptake
- **Defects in specific AA transporter systems can impair intestinal absorption or renal tubular reabsorption of the affected AAs.**

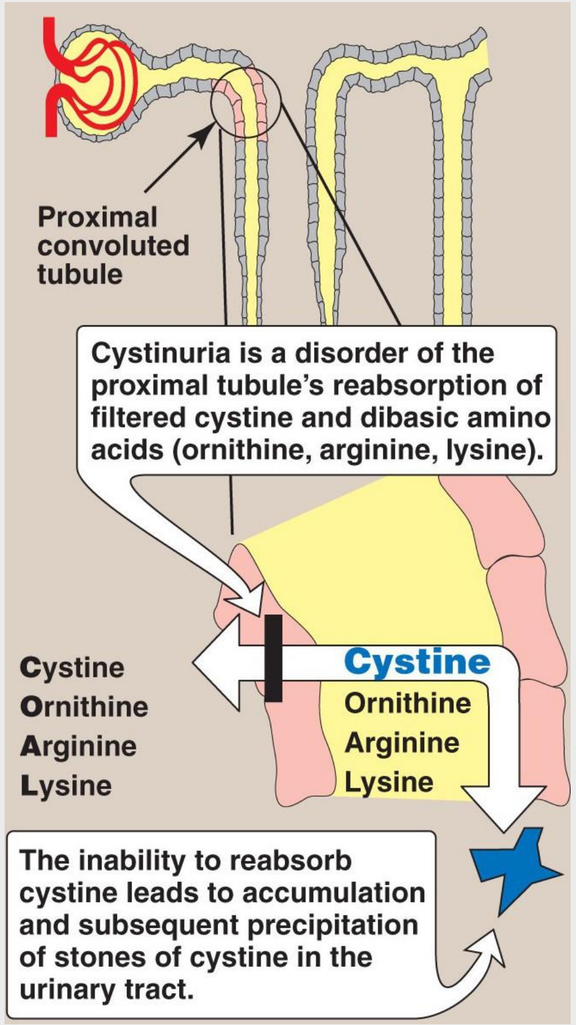
SYSTEM NAME	SODIUM-DEPENDENT?	SPECIFICITY	TISSUES EXPRESSED
A	Yes	Small and polar neutral amino acids (Ala, Ser, Gln, Gly, Pro, Cys, Asn, His, Met)	Many
ASC	Yes	Small amino acids (Ala, Ser, Cys)	Many
N	Yes	Gln, Asn, His	Liver, basolateral membrane of kidney
L	No	Branched and aromatic amino acids (His, Met, Leu, Ile, Val, Phe, Tyr, Trp)	Many[b]
$B^{0,+}$	Yes	Basic amino acids	Intestine (brush border)[c] and kidney
B^0	Yes	Zwitterionic amino acids (monoamino, monocarboxylic acid amino acids)	Intestine and kidney[d]
X_{AG}^-	Yes	Anionic amino acids (Asp, Glu)	Intestine (brush border) and kidney
Imino	Yes	Pro, hydroxyproline, Gly	Intestine (brush border) and kidney

Modified from: Table 35.1, Chapter 35: "Marks' Basic Med. Biochem.", 6e

Amino Acid Transport Defects

Clinical correlation

Cystinuria



Modified from: Figure 19.6, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e

Clinical correlation

Hartnup Disease

- Autosomal-recessive mutations in a **Na+-dependent neutral amino acid co-transporter (tryptophan)**
- Affected individuals may develop **pellagra-like symptoms** (photosensitivity, dementia) as a result of tryptophan deficiency.
- **Tryptophan is a precursor in niacin (vitamin B3) synthesis, required for NAD^+ and $NADP^+$ synthesis,** the mutation can result in niacin deficiency
- **Alternate routes for tryptophan absorption can partially compensate for the defect**
- **Patients usually require dietary niacin supplementation.**

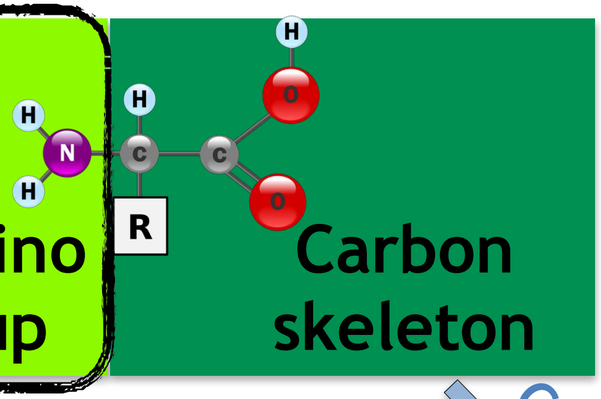
Nitrogen Metabolism: Step 1

Transamination (transfer of amino groups)

Oxidative deamination (liberation of amino groups as ammonia)

STEP 1

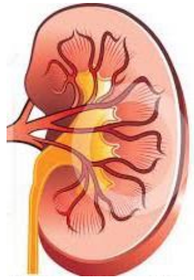
Removal of the α-amino group

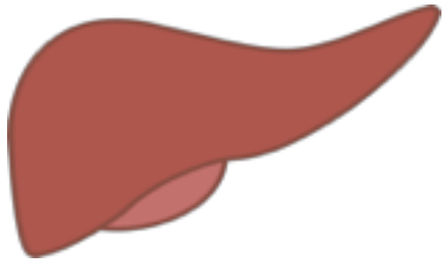


NH_3

urea

excretion



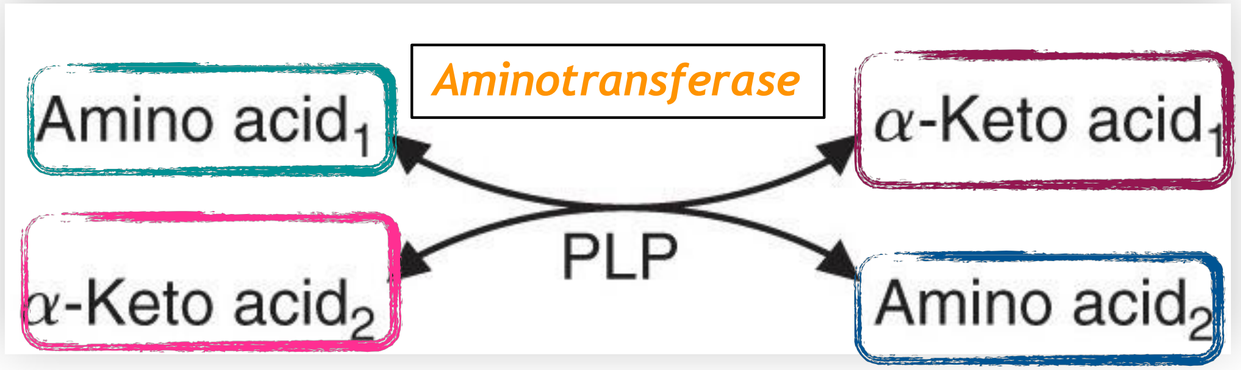


Anabolism

Catabolism

various metabolic pathways

Transamination



Transamination typically involves **α-ketoglutarate** as the amino group acceptor.

- The process is catalyzed by **aminotransferases** (transaminases).
- Allows amino nitrogen from different AAs to be collected in the form of glutamate, glutamate can subsequently release its amino group as ammonia through **Oxidative deamination**.

It's reversible and plays a central role in amino acid metabolism.

Lysine and threonine are exceptions and are often cited as non-transaminating amino acids.

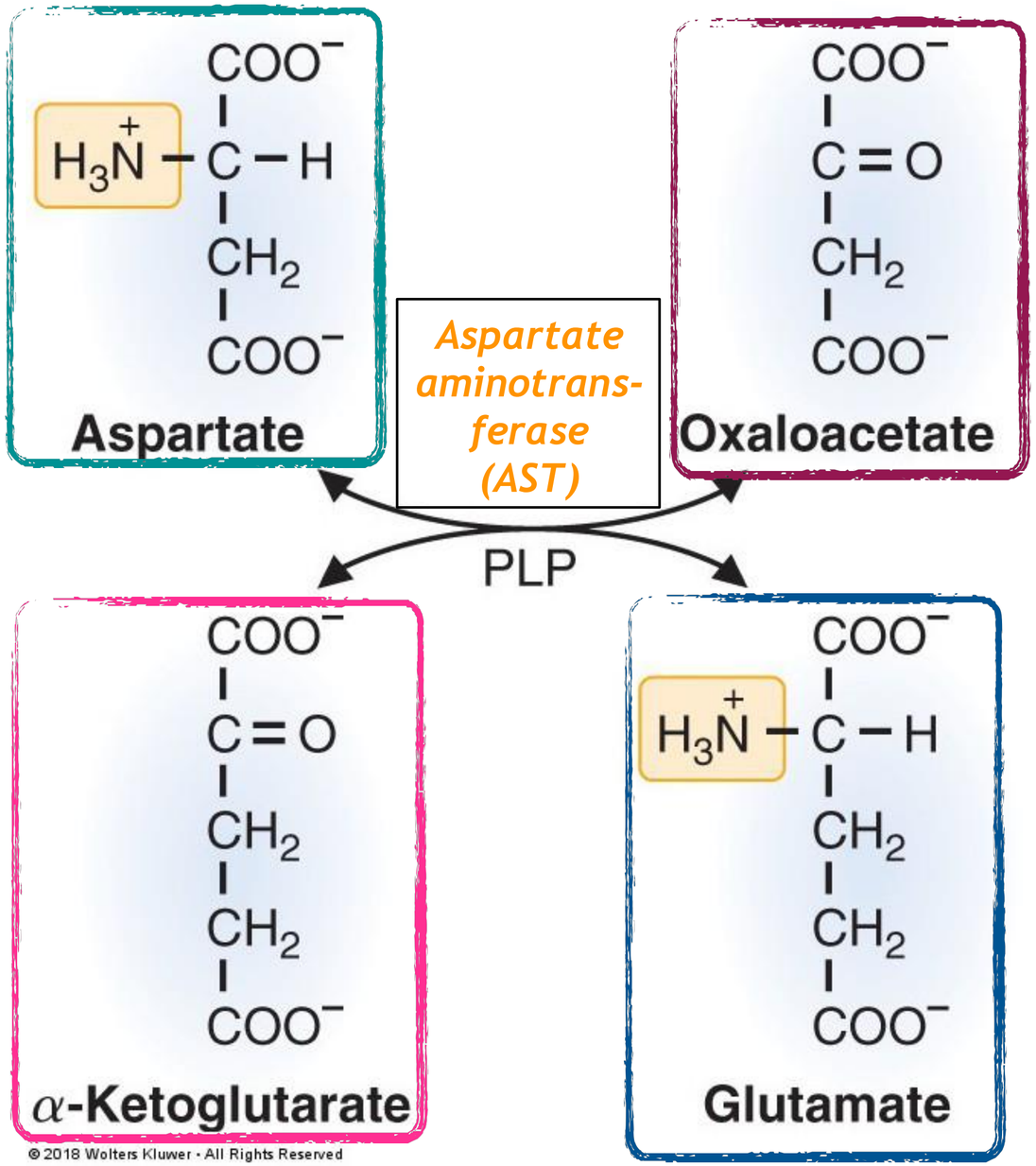
Modified from: Figure 36.3, Chapter 36: Marks' Basic Med. Biochem.", 6e

Transamination

Aminotransferases: Transfer an amino group from an amino acid to an α-keto acid.

ALT (Alanine Aminotransferase): This enzyme is found predominantly in the liver. While it exists in other tissues, it is considered a more specific indicator of liver damage than AST.

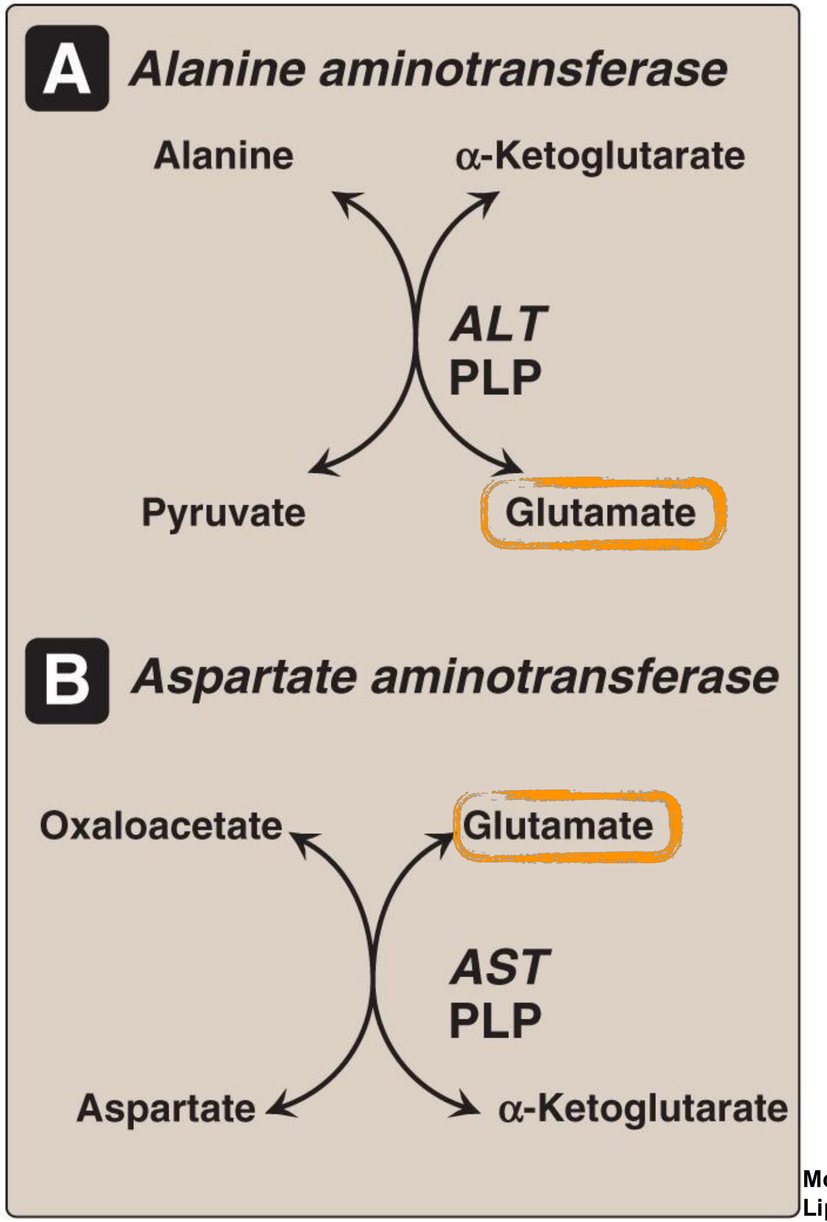
AST (Aspartate Aminotransferase): AST is found in high concentrations in the liver, heart, skeletal muscle, and kidneys. Because of its wider distribution, elevated AST levels can indicate damage to these other organs as well as the liver.



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Transamination



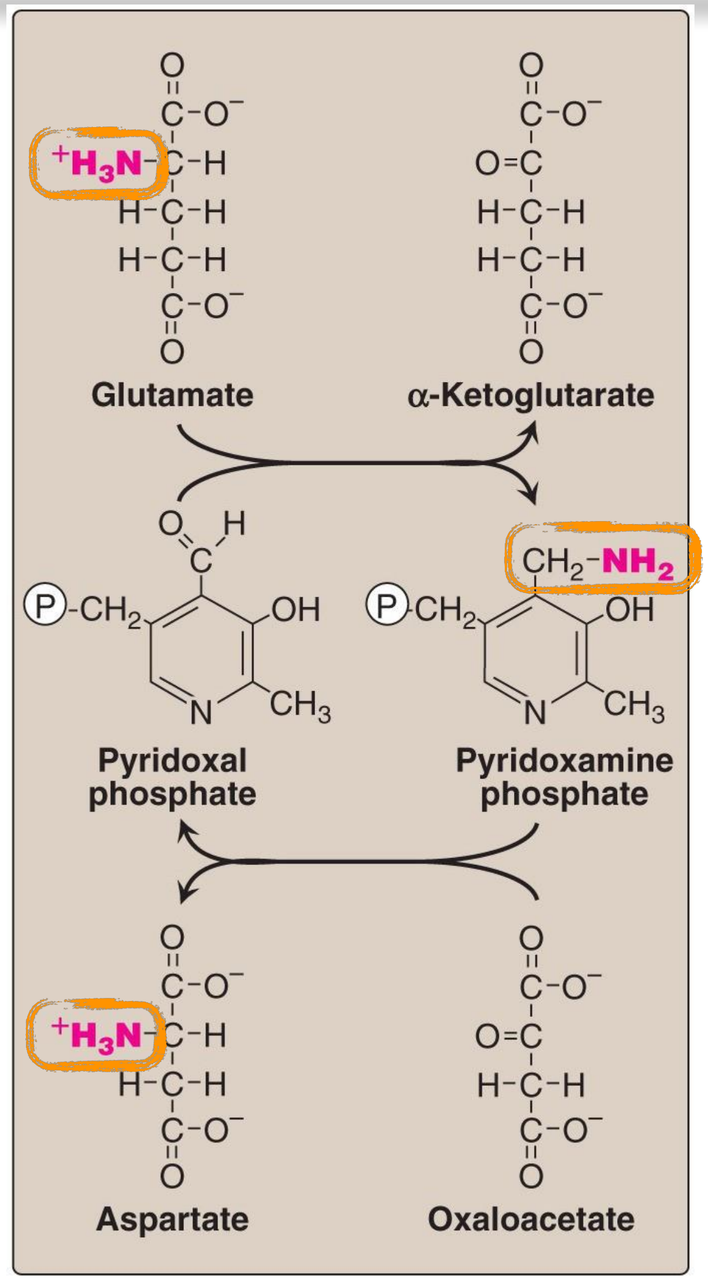
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Enzymes: *Aminotransferases (transaminases)*

- ✓ **Each aminotransferase recognizes one or a limited group of amino acid substrates.**
- ✓ **Widely distributed in the body tissues, with cytosolic and mitochondrial isoenzymes.**
- ✓ **All of them require the coenzyme pyridoxal phosphate (PLP, a vitamin B6 derivative) that participate in the amino group exchange.**
- ✓ **Reversible reactions** with constant equilibrium **near 1 which allow them to** function in both:
  - **AA degradation** through removal of α-amino groups.
  - **NEAA biosynthesis** through addition of amino groups to the carbon skeletons of α-keto acids.

Modified from: Figure 19.8, Chapter 19: Lippincott's Ills. Reviews: Biochem.", e

Transamination: Mechanism



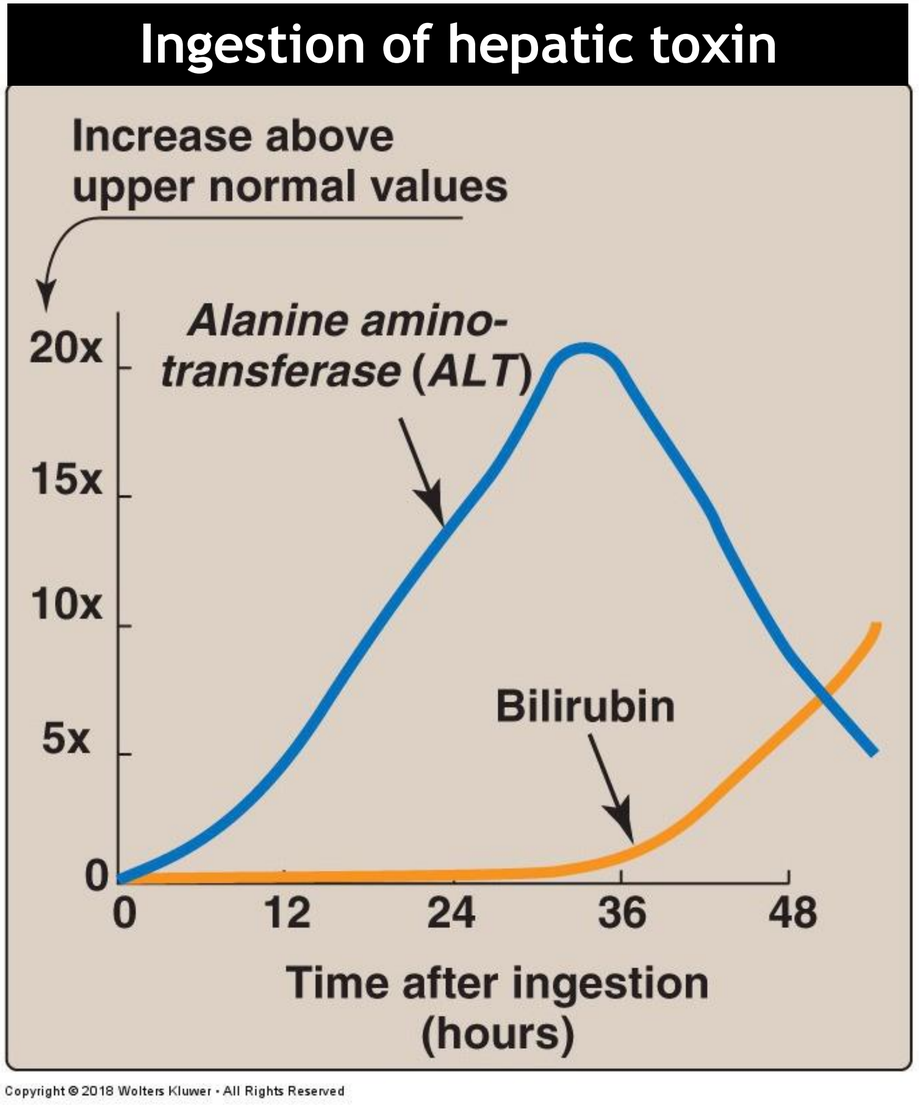
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Enzyme: *Aspartate aminotransferase (AST)*

- ✓Require the coenzyme pyridoxal phosphate (PLP, a vitamin B6 derivative), act as a temporary carrier for the amino group during the transamination reaction.
- ✓Reversible reaction
- ✓During amino acid metabolism, aminotransferases funnel amino groups to form glutamate
- ✓AST can operate in the reverse reaction, primarily transfers amino groups from glutamate to oxaloacetate, forming aspartate, which is used as a source of nitrogen in the urea cycle

Modified from: Figure 19.9, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

Aminotransferases in Clinical Diagnosis



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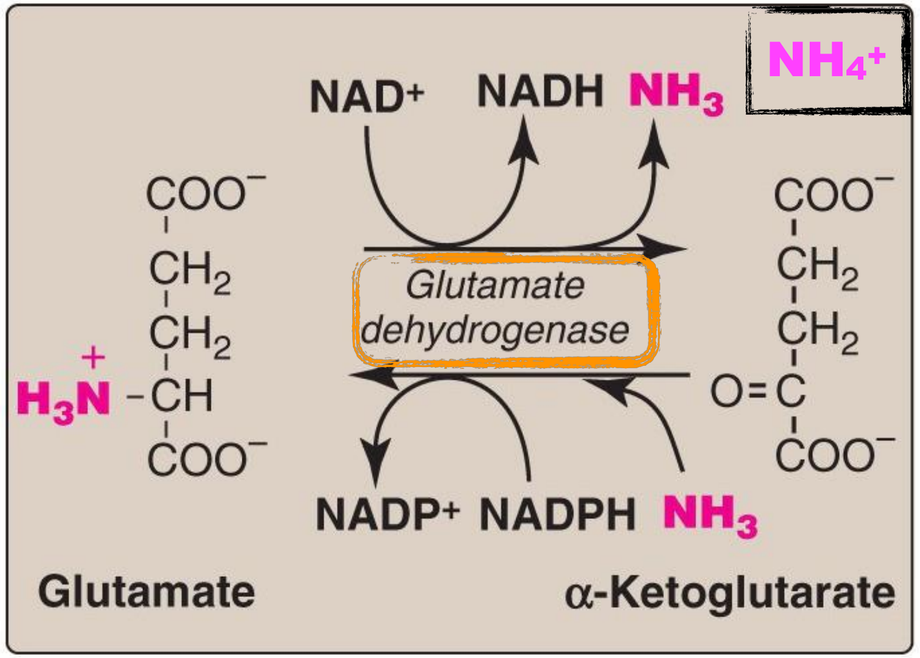
Hepatic diseases

- **Plasma AST and ALT are elevated**
- Very high in conditions that cause extensive cell necrosis (severe viral hepatitis, toxic injury, and prolonged circulatory collapse)
- **ALT is more specific than AST for hepatocellular injury, predominantly cytosolic.**
- **AST is also abundant in the liver, but widely distributed in other tissues, has both cytosolic and mitochondrial forms.**
- Serial measurements of AST and ALT (liver function tests) are often useful in determining the course of liver damage

Nonhepatic diseases

- **AST may also be elevated in following cardiac or skeletal muscle injury.**
- Clinical findings and additional tests help distinguish these conditions from hepatocellular injury.

Oxidative Deamination of Glutamate



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Modified from: Figure 19.11, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

NH_4^+ - ammonium ion is in aqueous solutions

NH_3 - crosses the membranes

Enzyme: *Glutamate dehydrogenase*

✓**Expressed at high levels in liver and kidney**

✓**Mitochondrial localization**

✓**It can use either one as a coenzyme:**

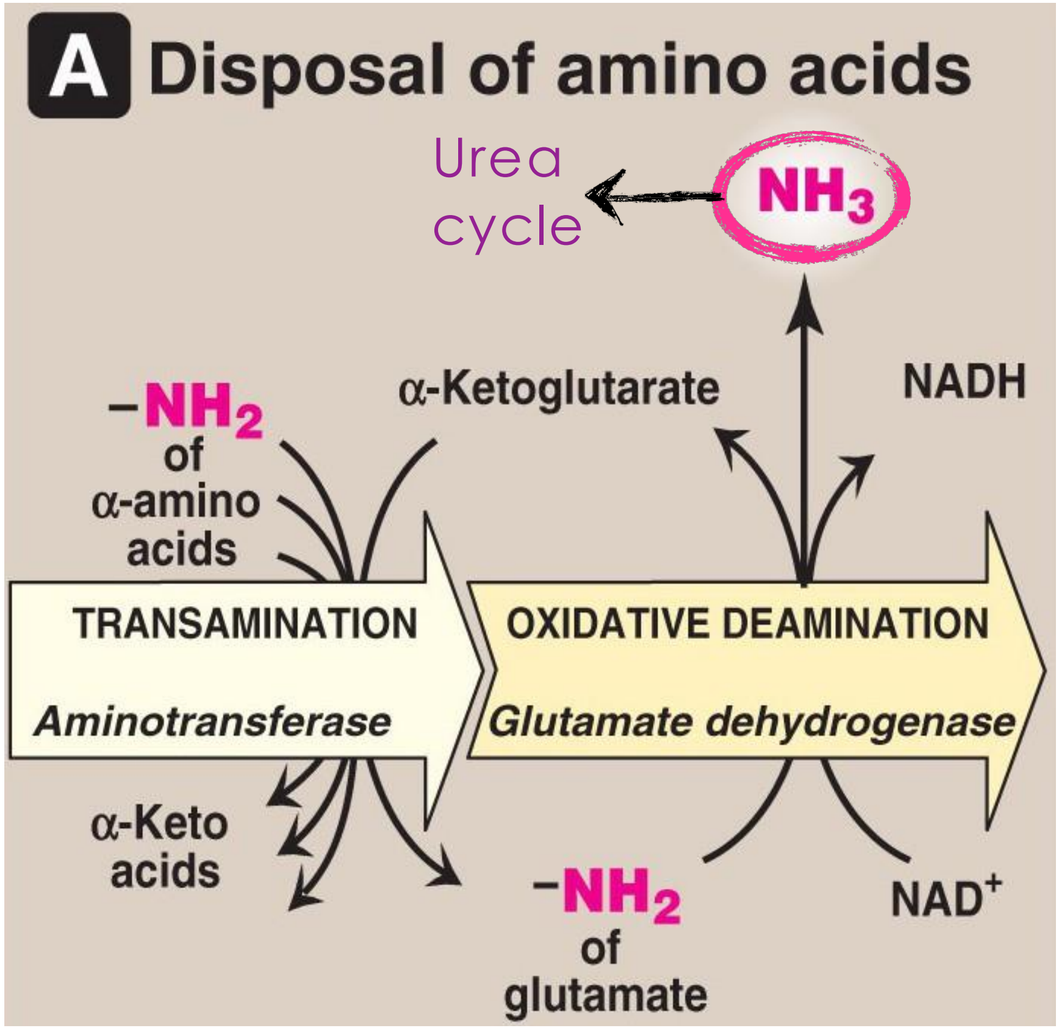
- **NAD^+ in oxidative deamination** – loss of NH_3 with simultaneous oxidation of the carbon skeleton
- **NADPH in reductive amination** – gain of NH_3 with reduction of the carbon skeleton

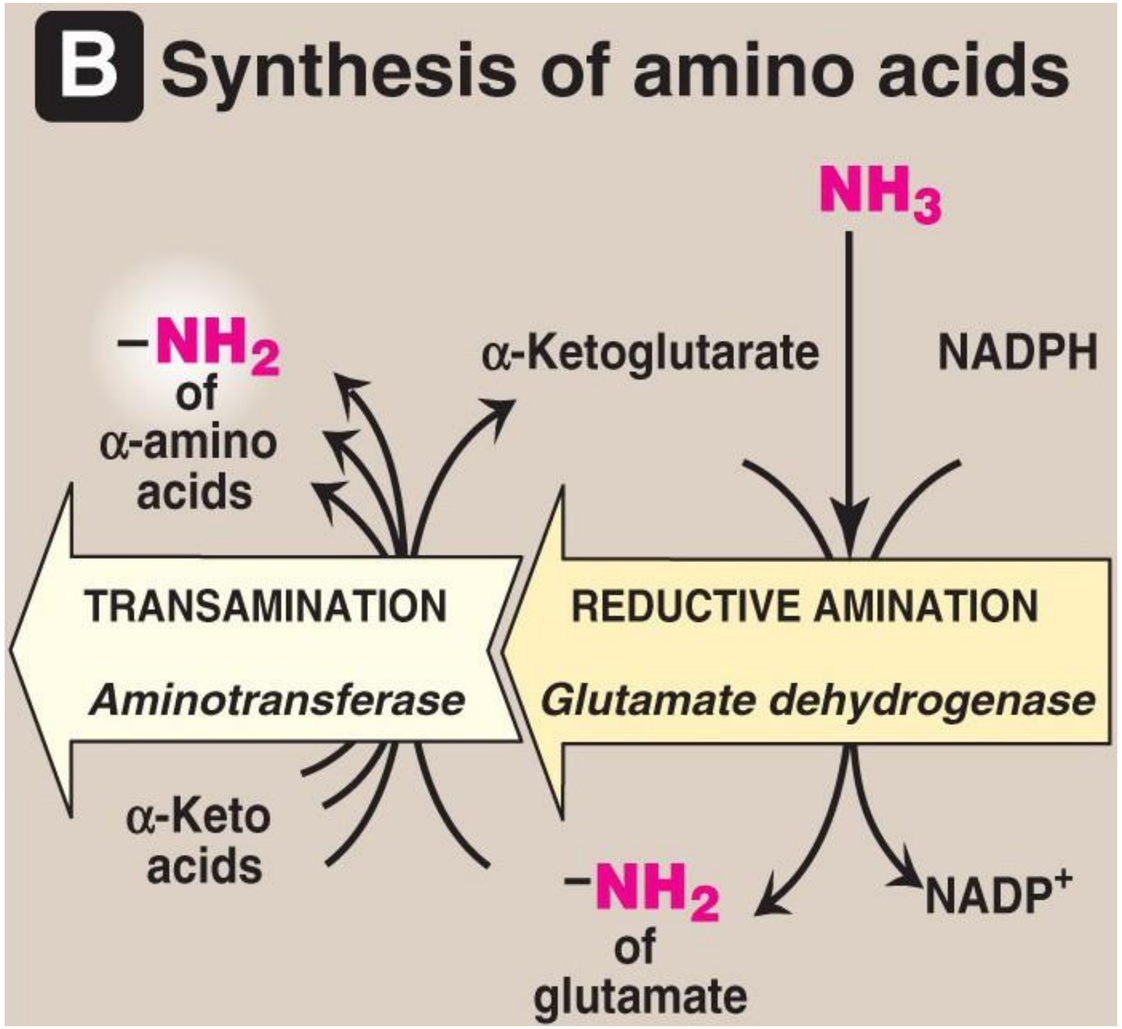
✓The direction depends on the relative concentrations of **glutamate, α-ketoglutarate, and NH_3** and the ratio of oxidized to reduced coenzymes

✓**Allosteric regulation:**

- **GTP - inhibitor**
- **GDP - activator**

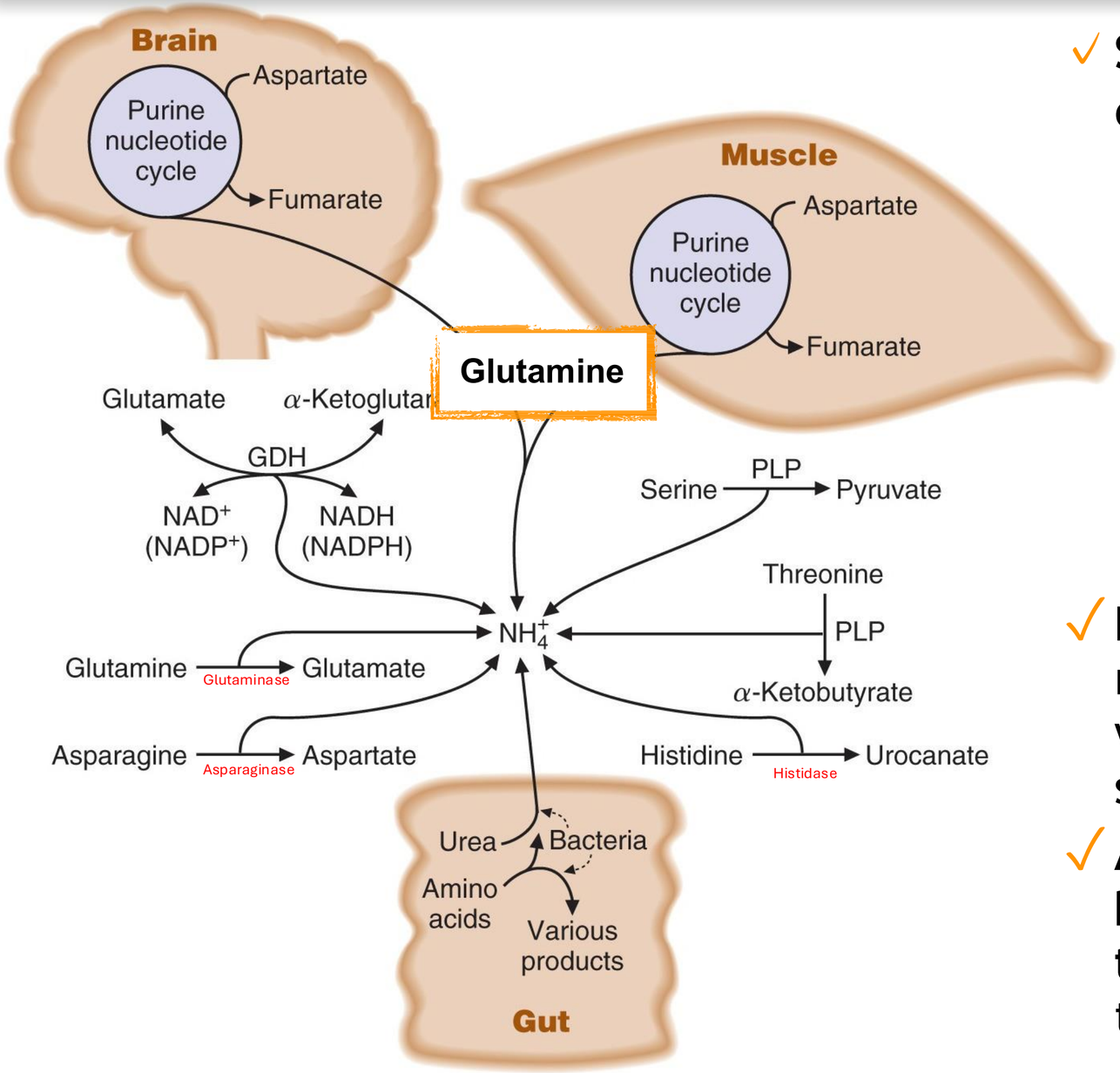
Role of Glutamate





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Other Sources of Ammonia

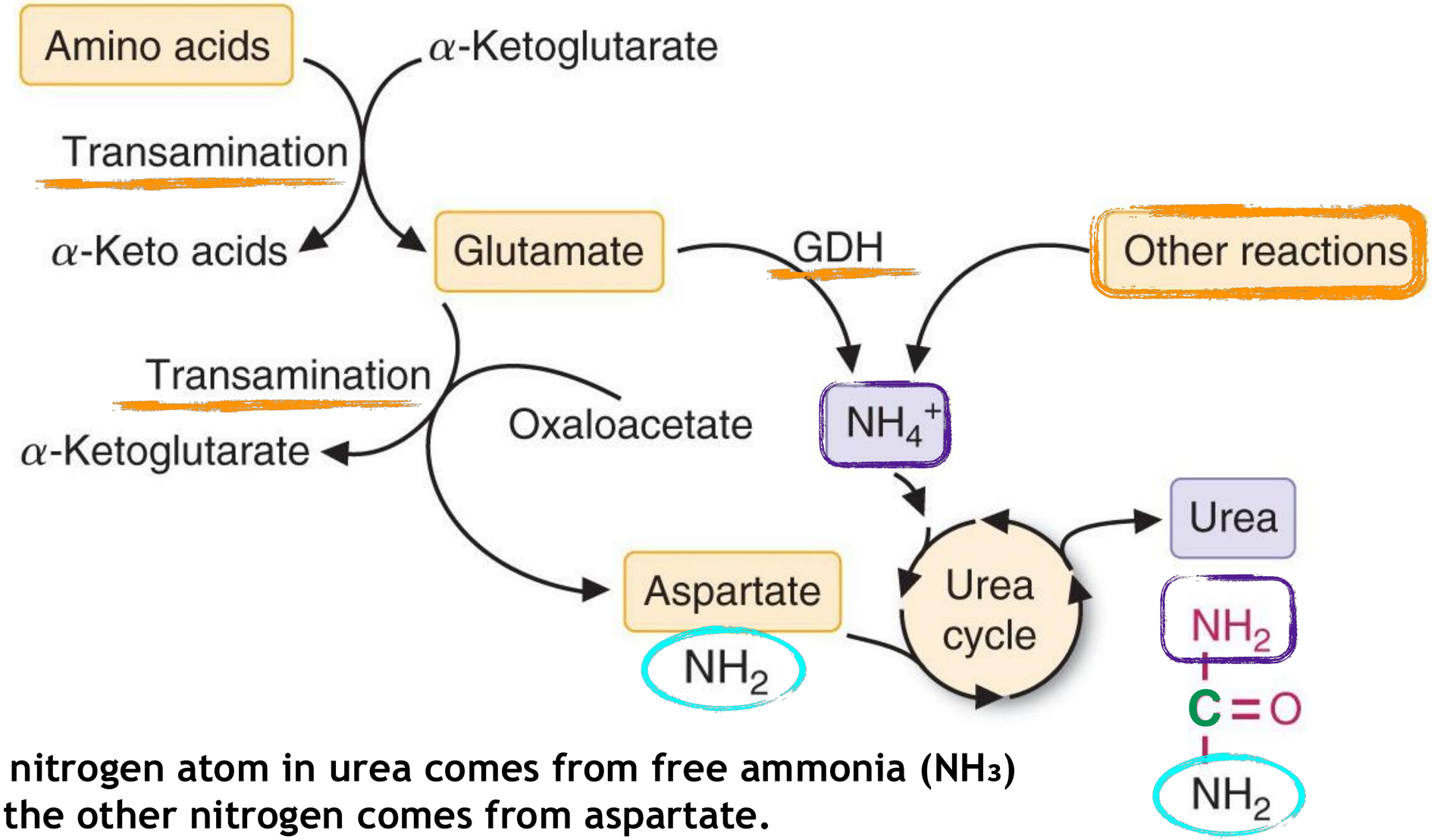


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- ✓ **Several amino acids release their nitrogen directly as NH_4^+:**
  - **Histidine** - direct deamination to form NH_4^+ and urocanate
  - **Serine and threonine** are directly deaminated by dehydration reactions that require PLP
  - **Glutamine and asparagine** R-group amides may be released as NH_4^+ by **deamidation**
  - The **glutaminase reaction is particularly important in the kidney** (pH balance)
- ✓ **In muscle and brain,** the purine nucleotide cycle releases NH4+ derived from amino acid nitrogen, which can be incorporated into glutamine for safe transport.
- ✓ **A significant amount of NH_4^+ is also produced by bacteria that live in the lumen of the intestinal tract.** It enters the hepatic portal vein and travels to the liver.

Modified from: Figure 36.4, Chapter 36: Marks' Basic Med. Biochem.", 6e

Role of Glutamate in N_2 Disposal

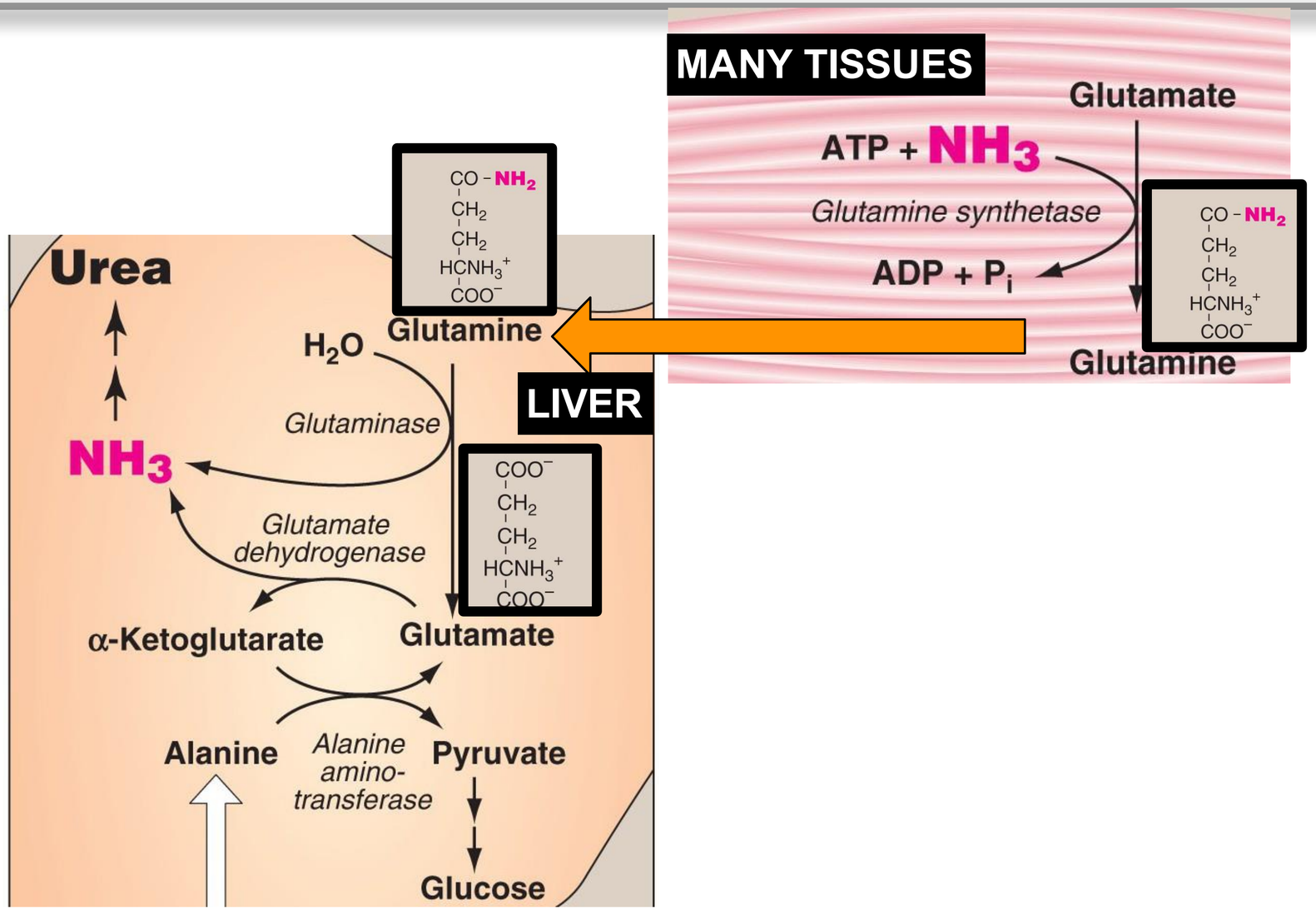


Modified from: Figure 36.7, Chapter 36: "Marks' Basic Med. Biochem.", 6e

One nitrogen atom in urea comes from free ammonia (NH_3) and the other nitrogen comes from aspartate.
Urea is transported in the blood to the kidneys, where it is excreted in the urine.

Transport of Ammonia to the Liver

MUST KNOW



As glutamine:

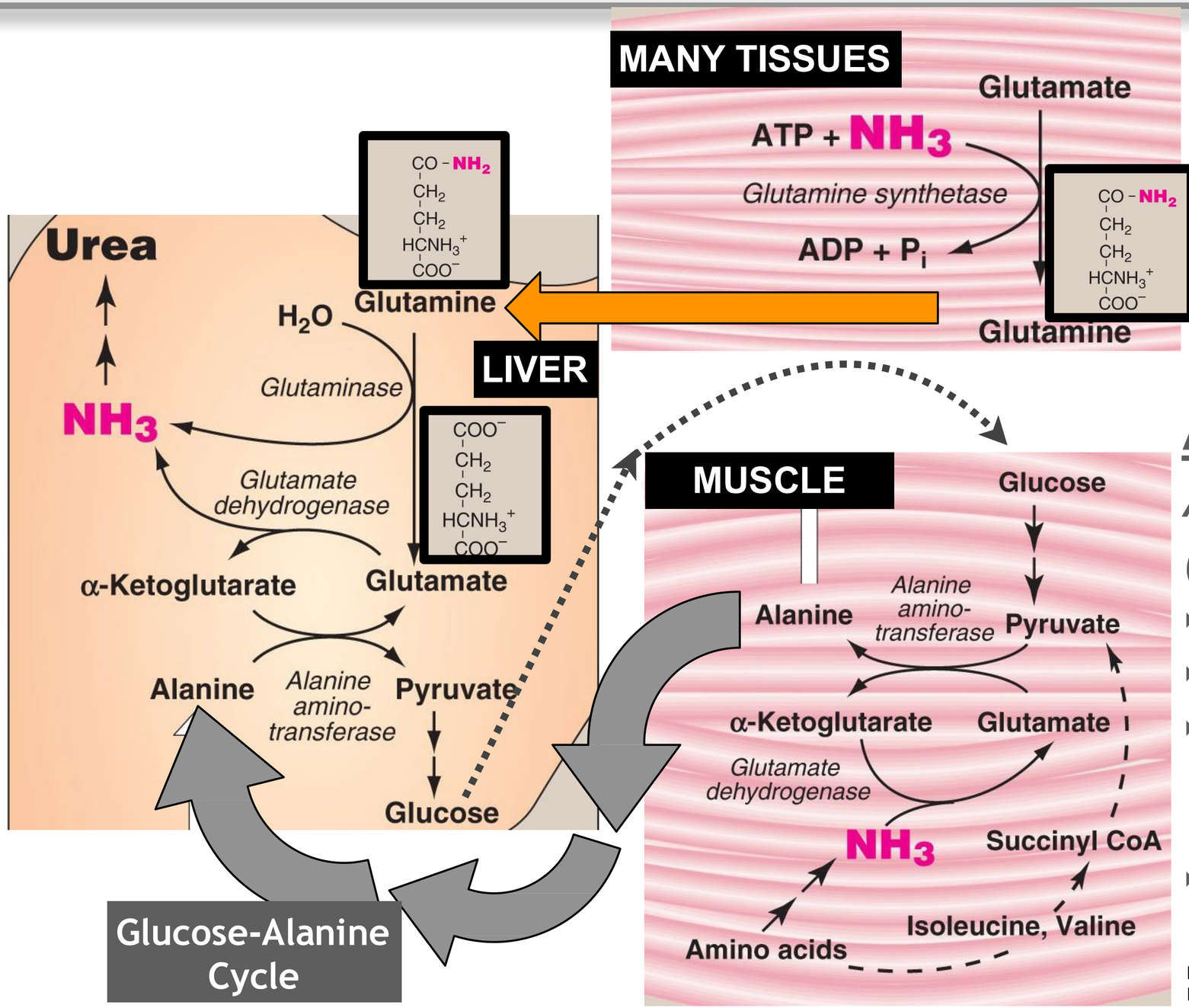
Glutamine synthetase

- free NH_3 is added to glutamate to make glutamine

Modified from: Figure 19.13, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

Transport of Ammonia to the Liver

MUST KNOW



As glutamine:

Glutamine synthetase

- free NH_3 is added to glutamate to make glutamine
- In all organs and has the highest concentration of all AAs in the circulation

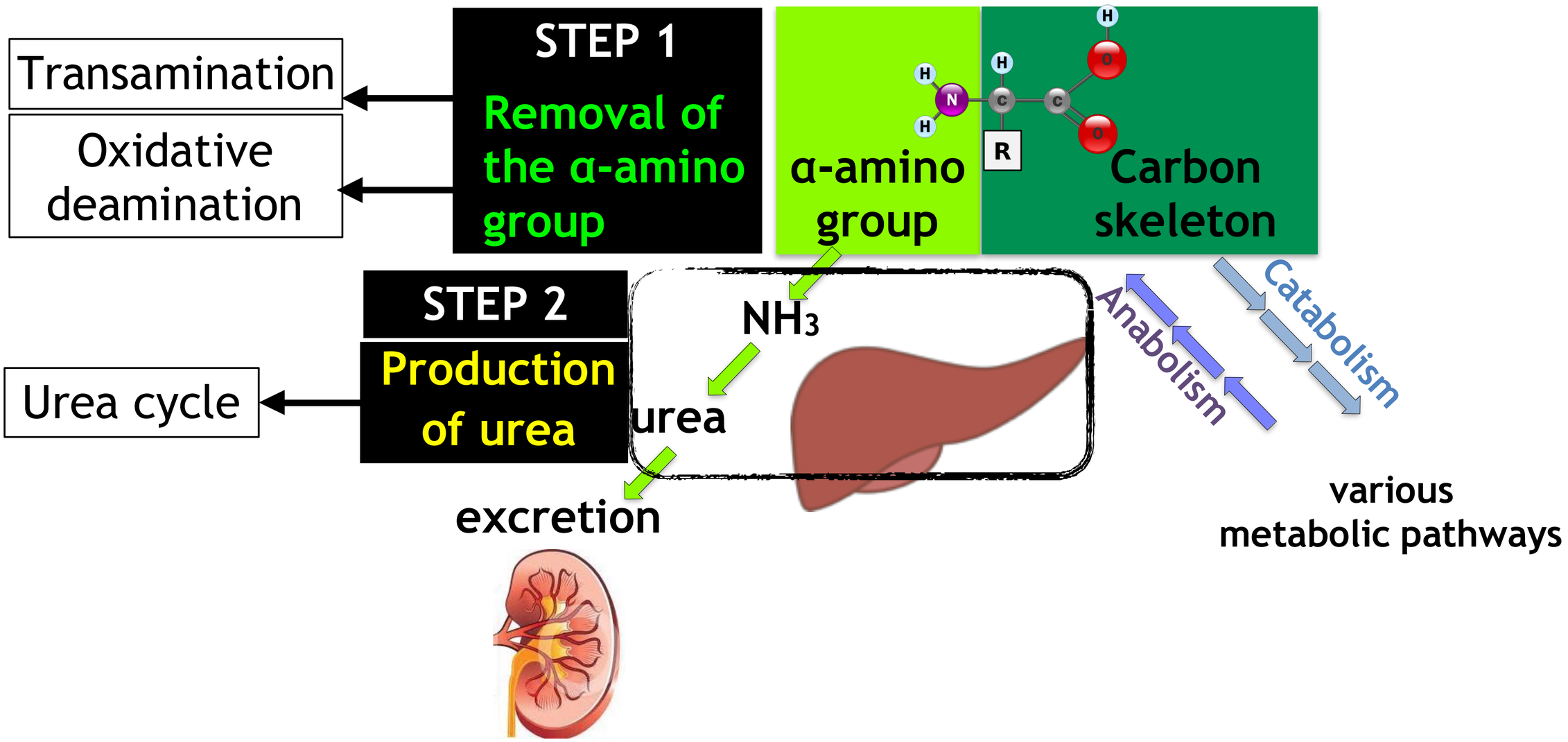
As alanine:

Alanine aminotransferase (ALT)

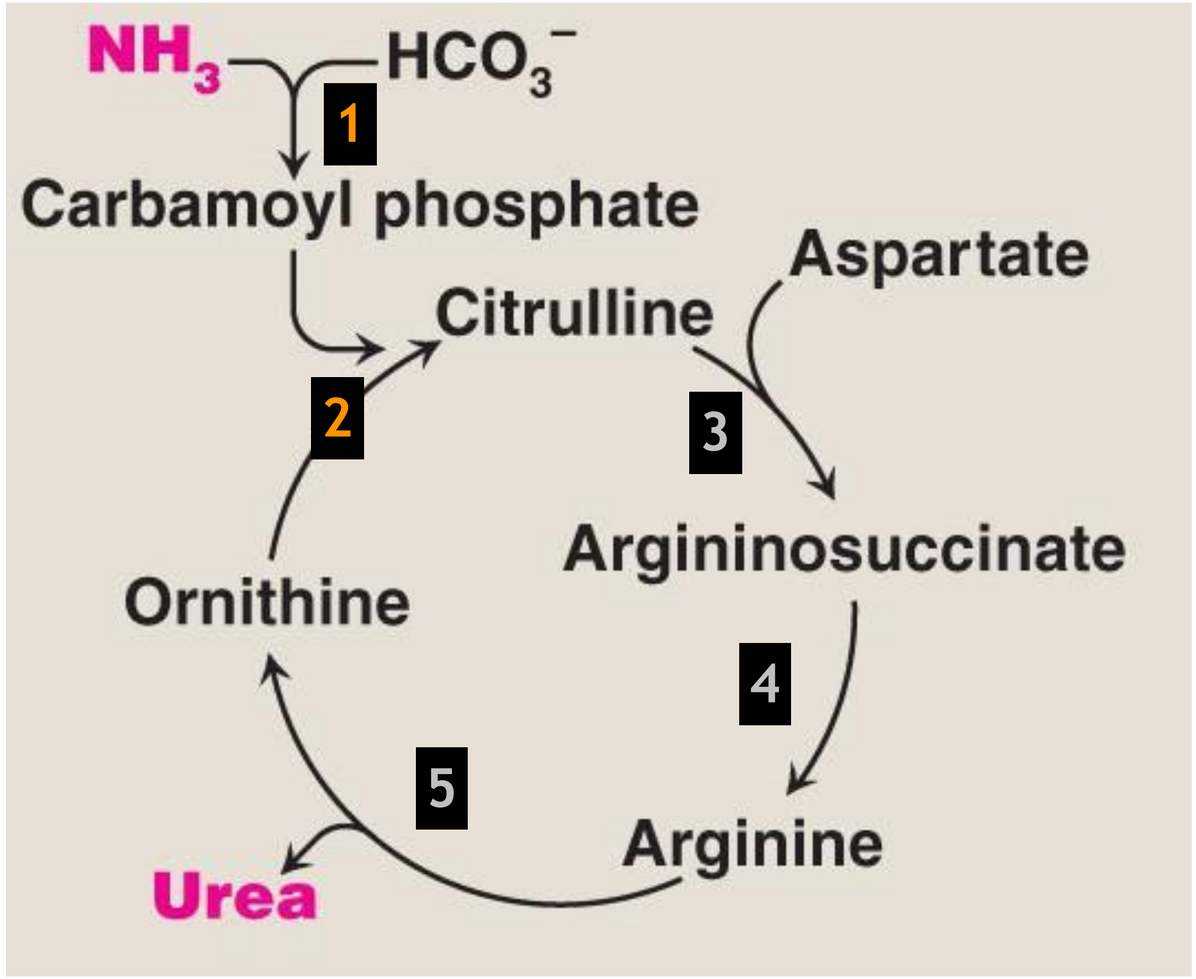
- Transamination of pyruvate
- Mainly in muscle cells
- In the liver, Ala is converted to pyruvate, again by transamination: the "glucose—alanine" cycle.
- In the muscle

Modified from: Figure 19.13, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e

Nitrogen Metabolism: Step 2



The Urea Cycle Overview



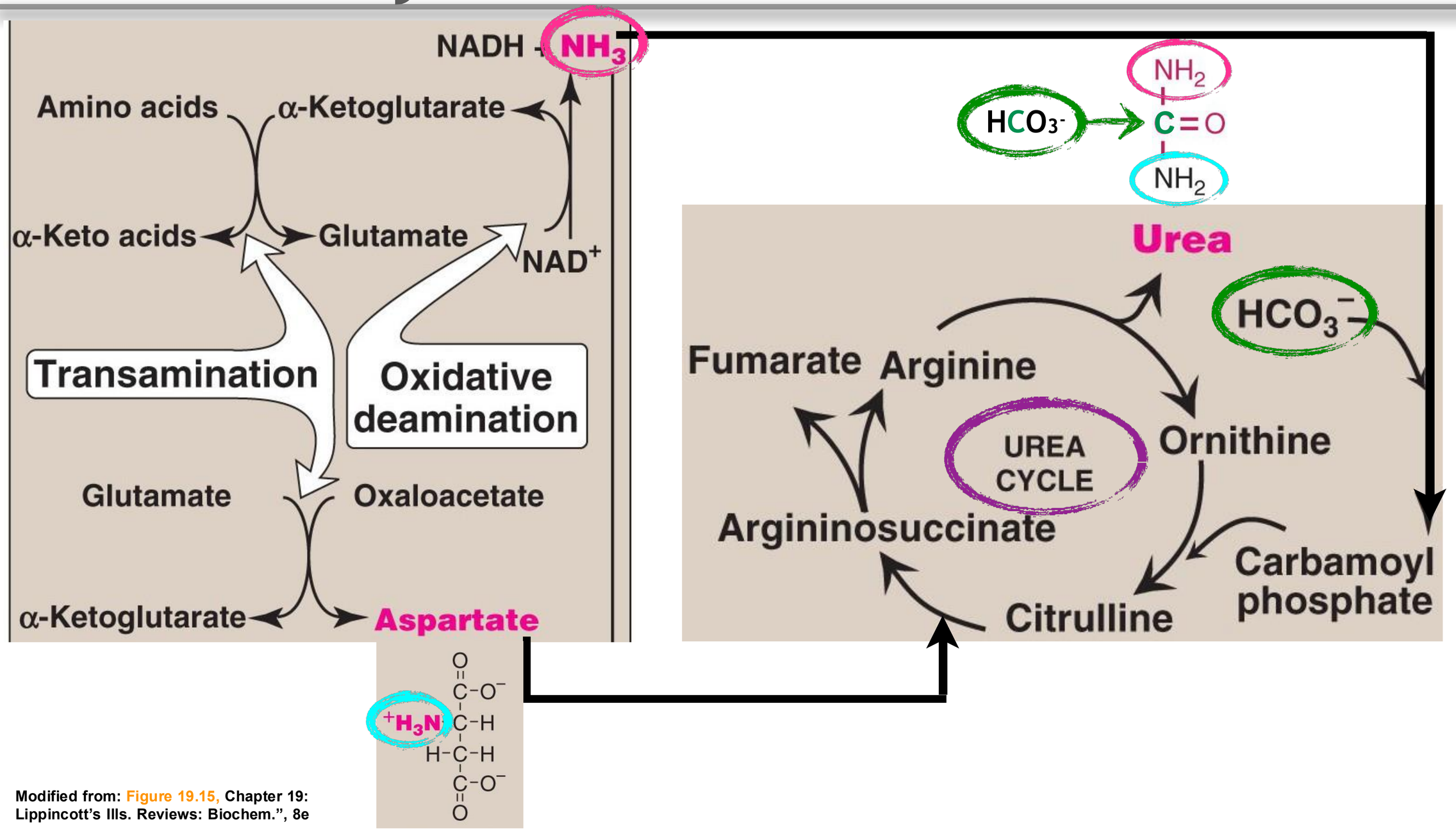
Modified from: Figure 19.1, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

- ✓Names: Urea cycle, the **ornithine cycle**
- ✓ A cycle of biochemical reactions occurring in that **produces urea from ammonia** (NH3)
- ✓Historically, this cycle was the first metabolic cycle discovered (Hans Krebs and Kurt Henseleit, 1932)
- ✓In mammals it **takes place primarily in the liver, and to a lesser extent in the kidney**

Consist of 5 reactions:

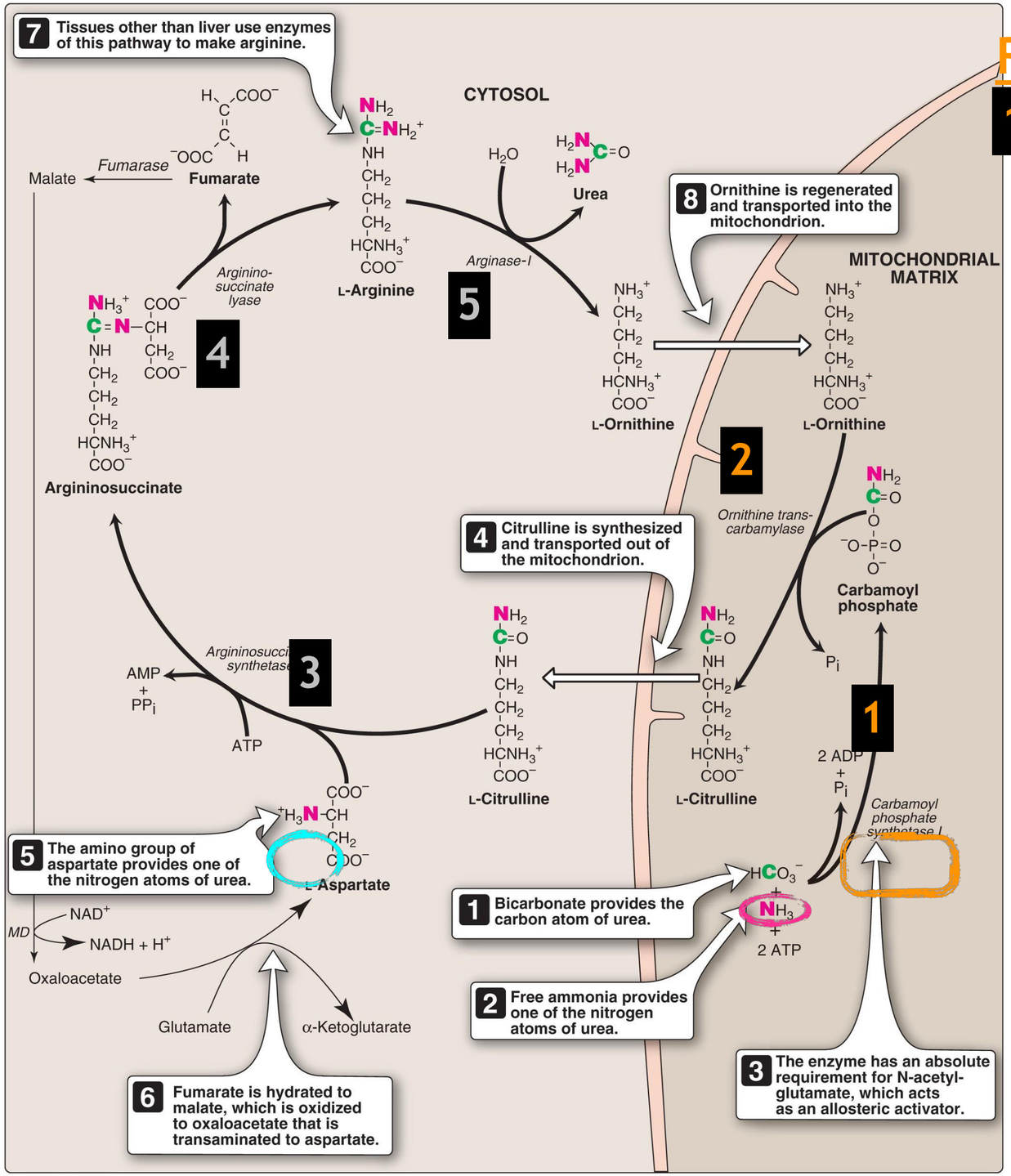
- 2 mitochondrial
- 3 cytosolic

The Urea Cycle Overview



Modified from: Figure 19.15, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

Rate-limiting regulated step:

1

Enzyme: *Carbamoyl Phosphate Synthetase I (CPS-I)*

Location: Mitochondrial Matrix

Reactants: Ammonia, bicarbonate, 2 ATP molecules

Product: Carbamoyl Phosphate

Regulation:

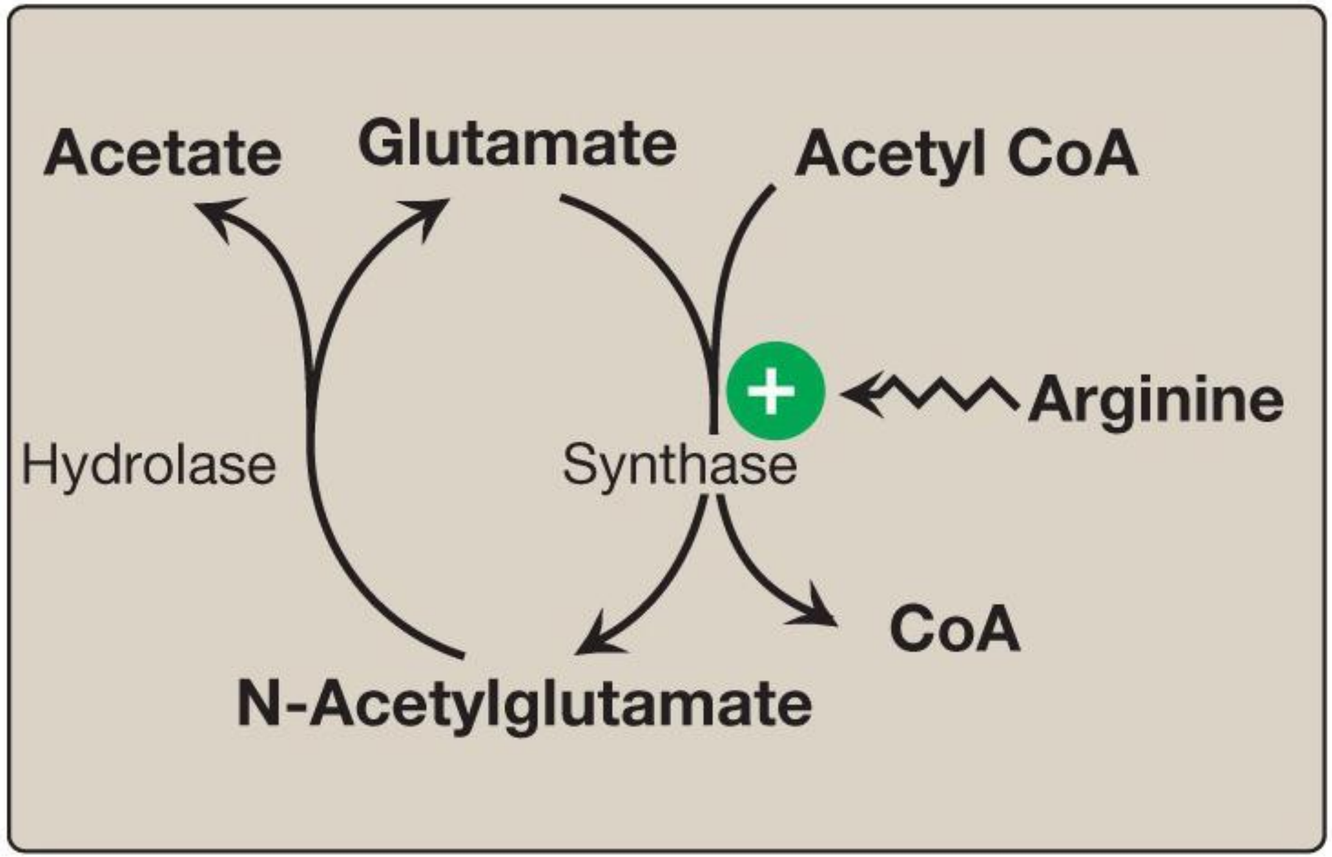
Short term:

- By substrate availability
- Allosteric activation by N-Acetylglutamate (NAG)

Long term:

- Enzyme induction

N-Acetylglutamate

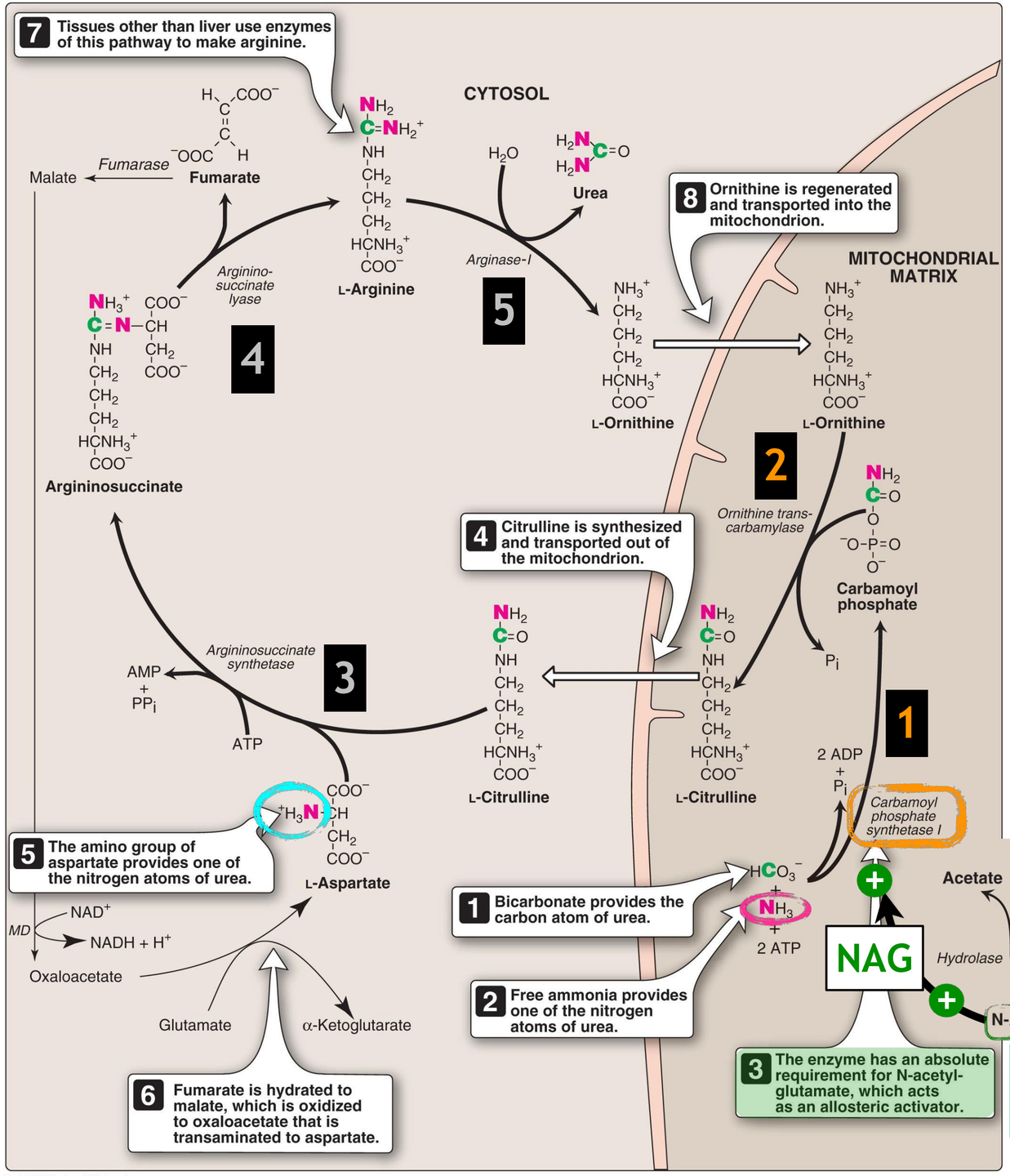


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Chapter 19: Lippincott's Illus. Reviews: Biochem., 9e

- N-acetylglutamate (NAG) is synthesized from the precursors L-glutamate and acetyl-CoA by N-acetylglutamate synthase (NAGS).
- NAG is an essential activator for CPS I, the rate-limiting step in the urea cycle.

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

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Rate-limiting regulated step:

1 **Enzyme:** *Carbamoyl Phosphate Synthetase I (CPSI)*

Regulation:

- **Short term:**

  Substrate availability and allosteric activation by NAG

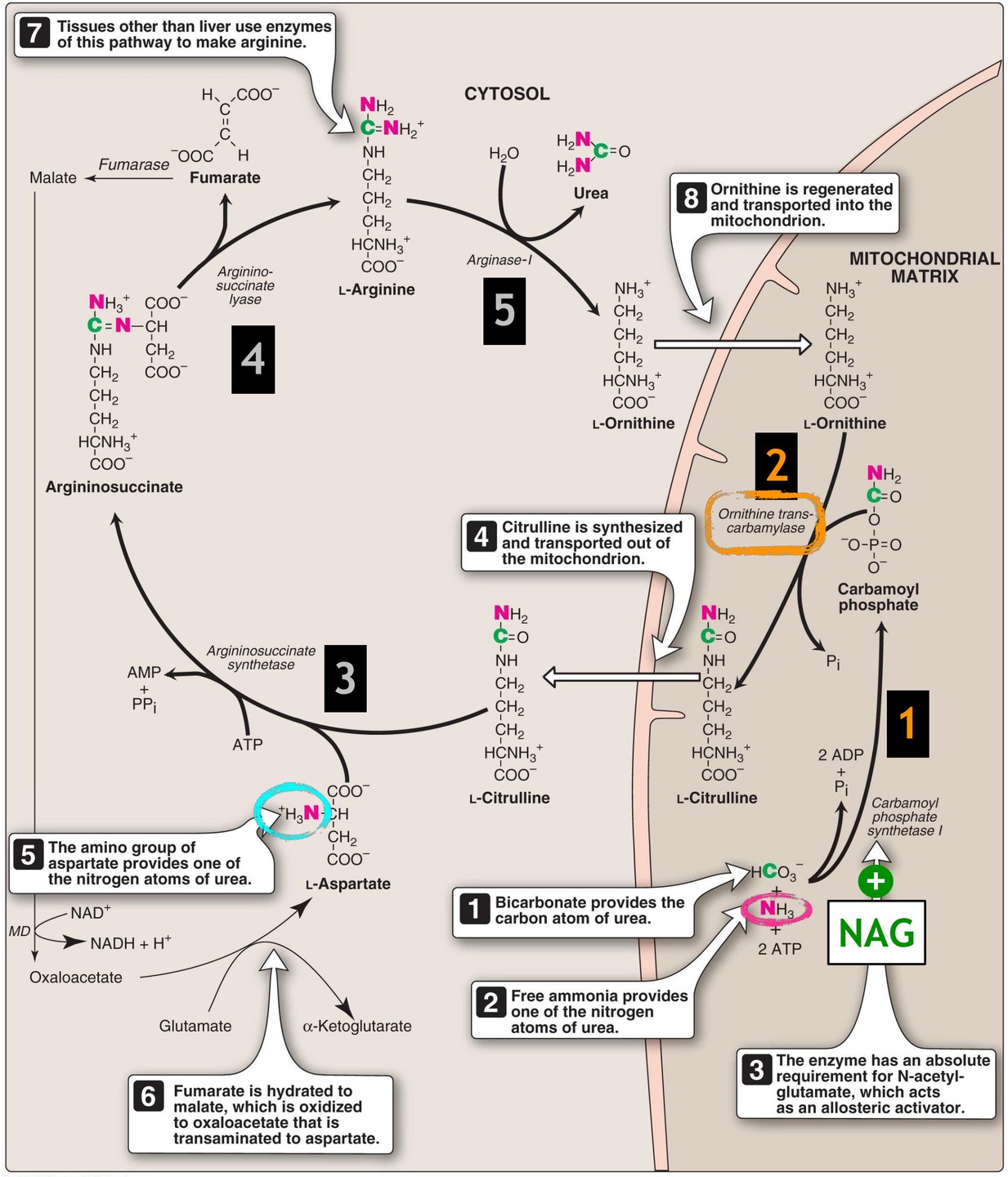
- **Long term:**

  A sustained high-protein diet increases transcription and expression of urea-cycle enzymes, allowing greater nitrogen disposal.

Urea cycle will be stimulated by a high protein meal!

A high protein diet will increase the transcription and expression of urea-cycle enzymes

The Urea Cycle



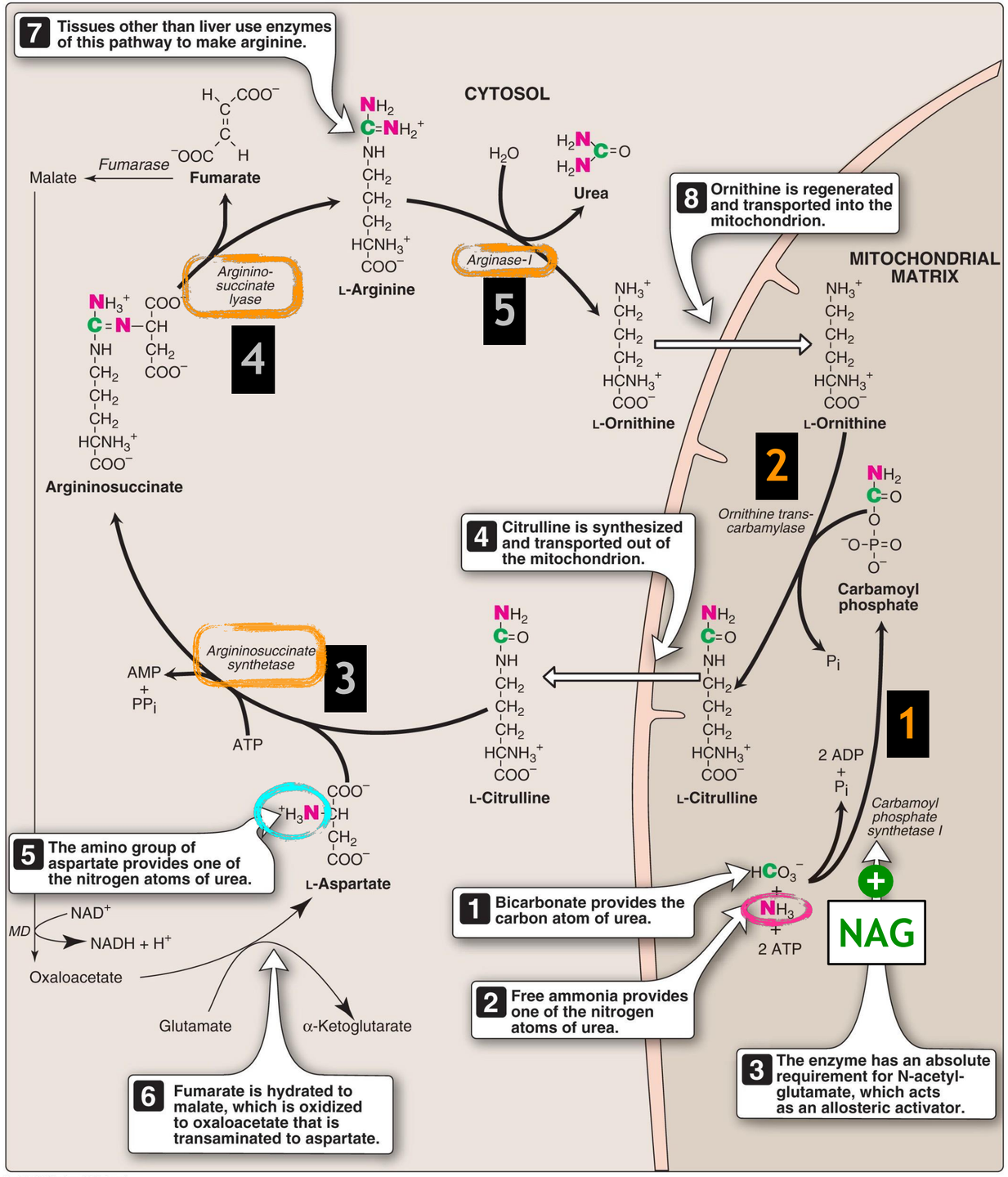
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2 Enzyme: *Ornithine transcarbamoylase (OTC)*

- In the mitochondria
- takes ornithine and carbomyl phosphate and makes citrulline
- Citrulline is transported to the cytosol
- A deficiency in this enzyme is the most common urea cycle defect (X-linked)

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

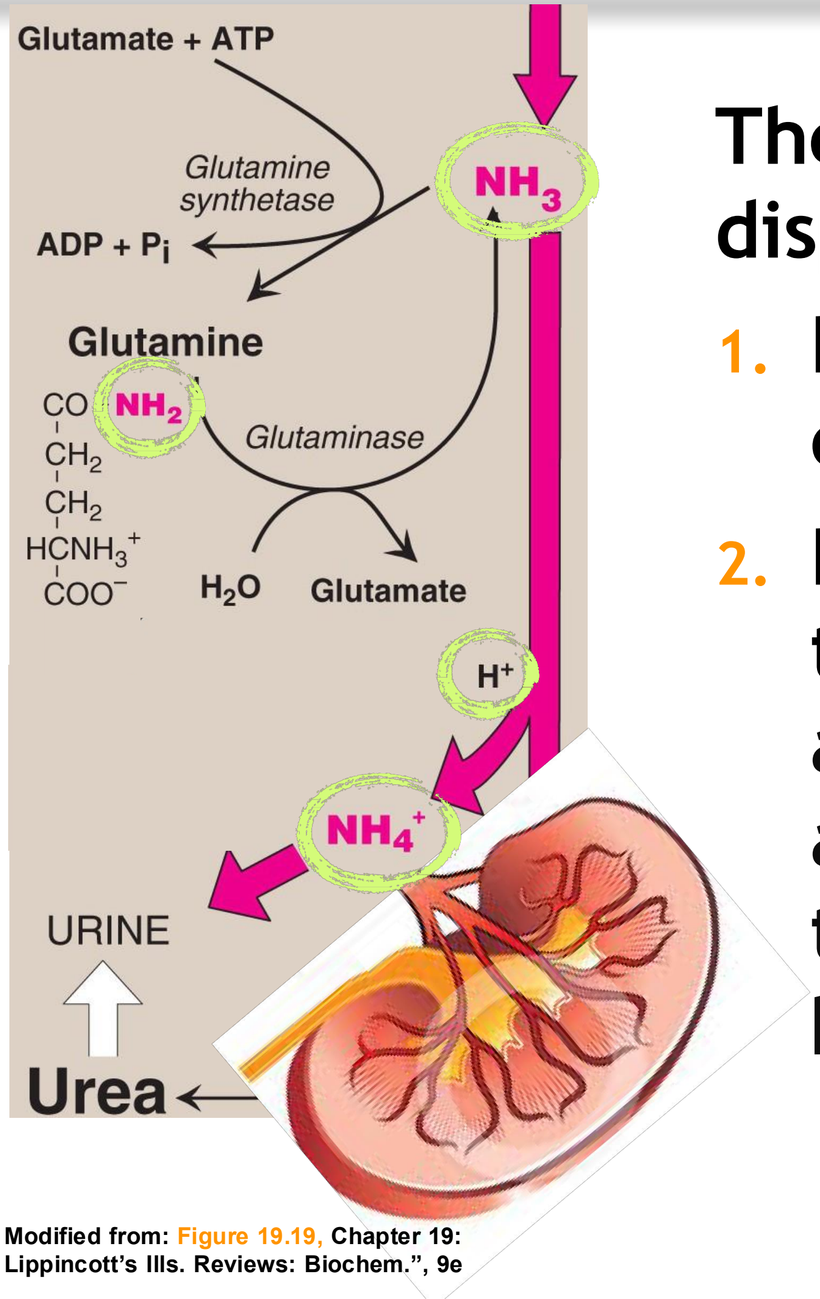
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3 4 5

- **All steps are in the cytosol**
- The second amino group which is added, **comes from** aspartate
- There are **common intermediates with the TCA cycle** (malate, fumarate, and oxaloacetate)
- **Enzyme:** *Arginase* releases **free urea** which is **transported into the bloodstream**, and filtered by the kidneys, where it will be **excreted in the urine.**
- Following the completion of the cycle, **ornithine is regenerated, and transported back into the mitochondria to be reused**

REFERENCE ADULT LABORATORY VALUES (NBOME)
Urea nitrogen, serum (BUN): 11-23 mg/dL

Role of Kidneys in Nitrogen Disposal

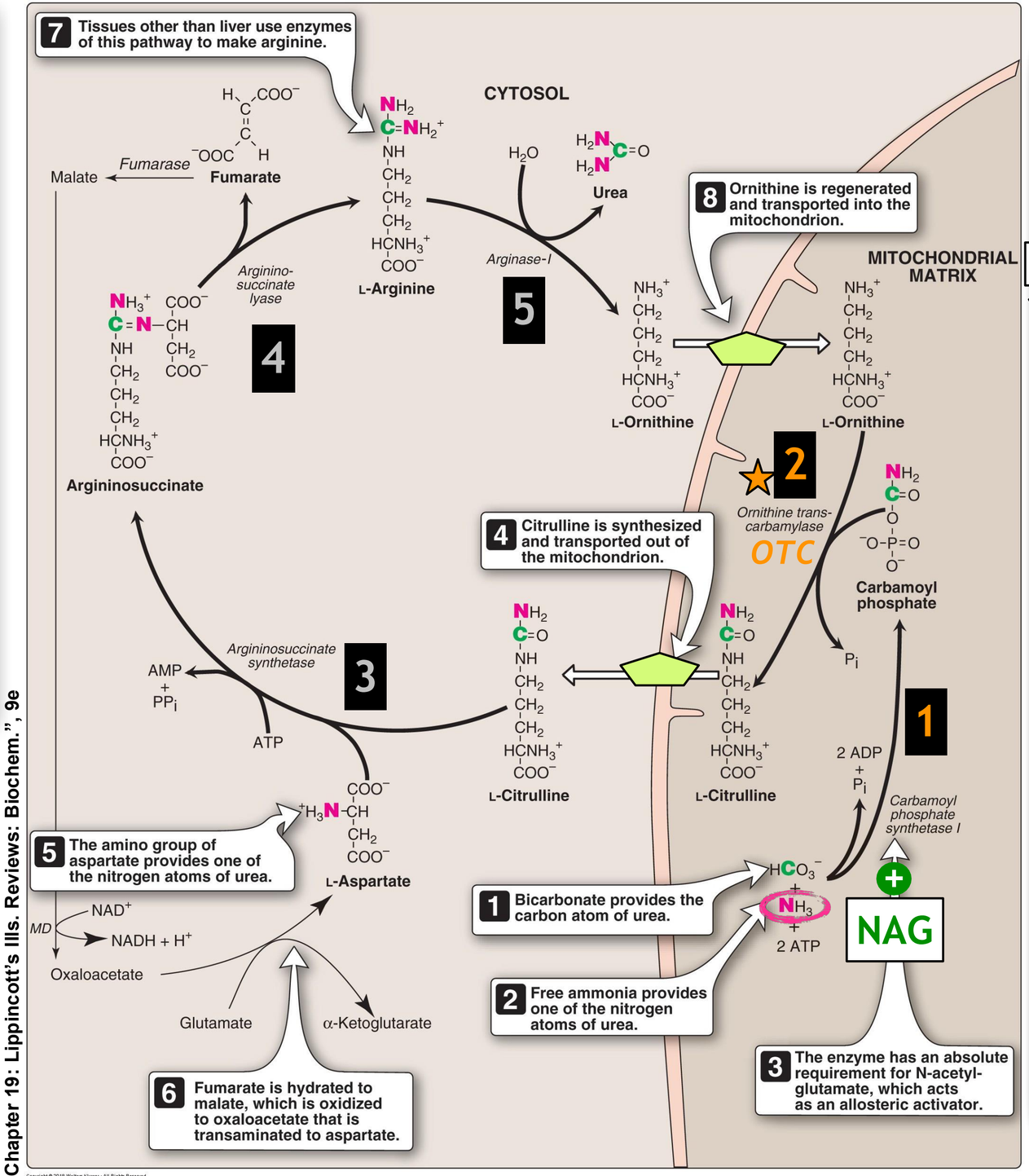


The kidneys have two distinct roles in nitrogen disposal:

1. Removal of urea produced by the liver from circulation.
2. Production of ammonia from glutamine through the action of glutaminase. This ammonia accepts a proton (H^+) to form ammonium ($NH4^+$), which is excreted in the urine, helping maintain acid-base balance.

Modified from: Figure 19.19, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 9e

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Clinical correlation

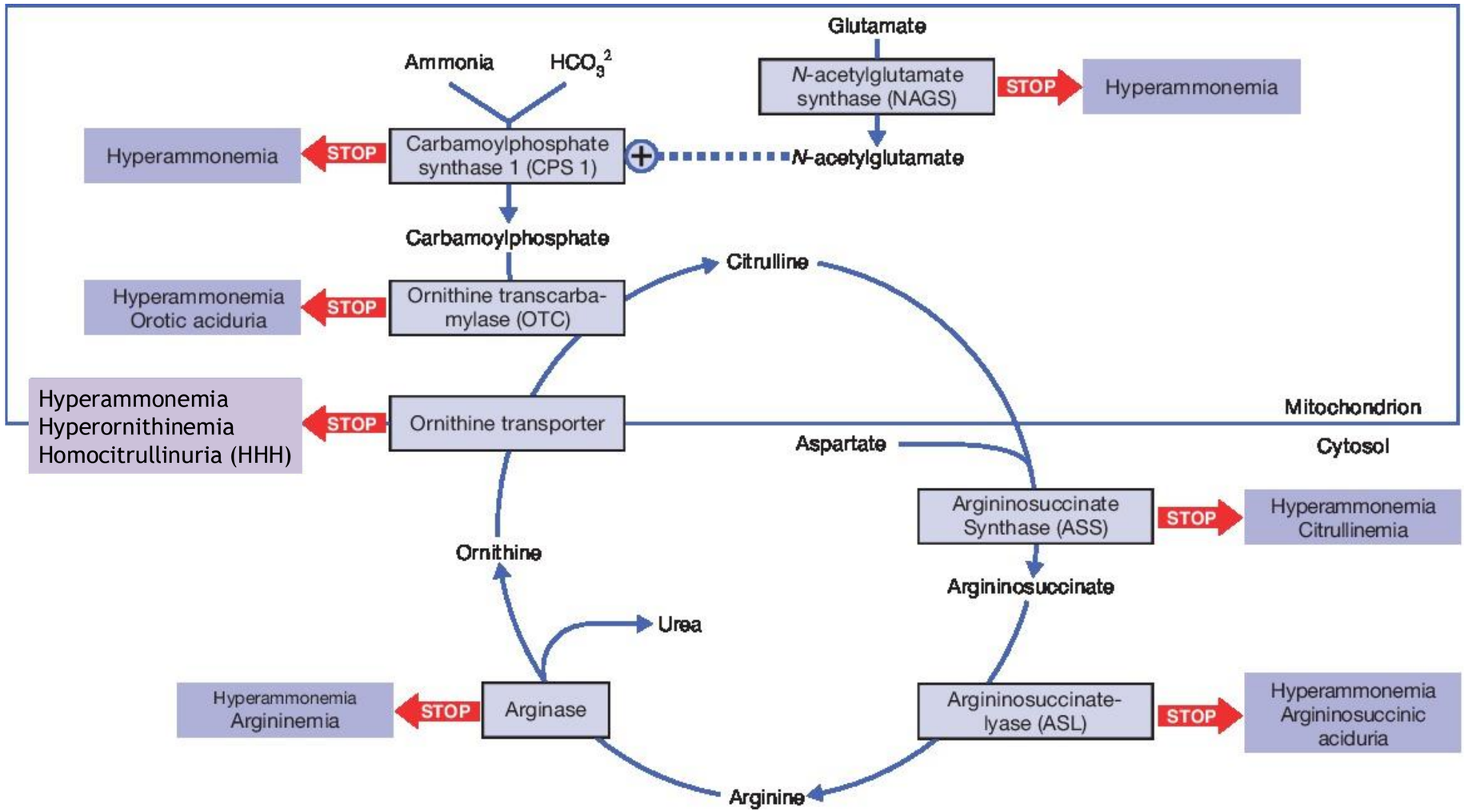
Deficiencies of Urea Cycle Enzymes

	Affected enzyme	Inheritance	Elevated in urine	Clinical presentation
1	Carbamoyl phosphate synthetase I	AR	Ornithine	Neonatal encephalopathy
NAG	*N* -acetylglutamate synthase	AR	Ornithine	Neonatal encephalopathy
	Ornithine translocase	AR	Ornithine, homocitrulline	Encephalopathy, liver dysfunction
2	Ornithine transcarbamoylase	XR	Ornithine	Neonatal encephalopathy
3	Argininosuccinate synthase	AR	Citrulline	Neonatal encephalopathy
4	Argininosuccinate lyase	AR	Argininosuccinate	Neonatal encephalopathy; tufted, friable hair
5	Arginase	AR	Arginine, cystine, lysine	Childhood protein intolerance, encephalopathy

Modified from: Table 28.1, Chapter 28: AA Metabolism; "Principles of Medical Biochem." 4e, 2017 by Elsevier

- ✓ *Generally cause hyperammonemia (neurotoxicity), severity depends on deficient enzyme.*
- ✓ *Urea-cycle intermediates may accumulate upstream the defective enzyme*
- ✓ *This prevents the a-ketoglutarate regeneration by removal of nitrogen from glutamine and free ammonia*
- ✓ *Defects in any enzyme lead to elevated glutamine and ammonia levels in the blood*
- ✓ *The extent of the elevation depends on which enzyme is defective*

Urea Cycle Disorders and Their Characteristic Biochemical Findings

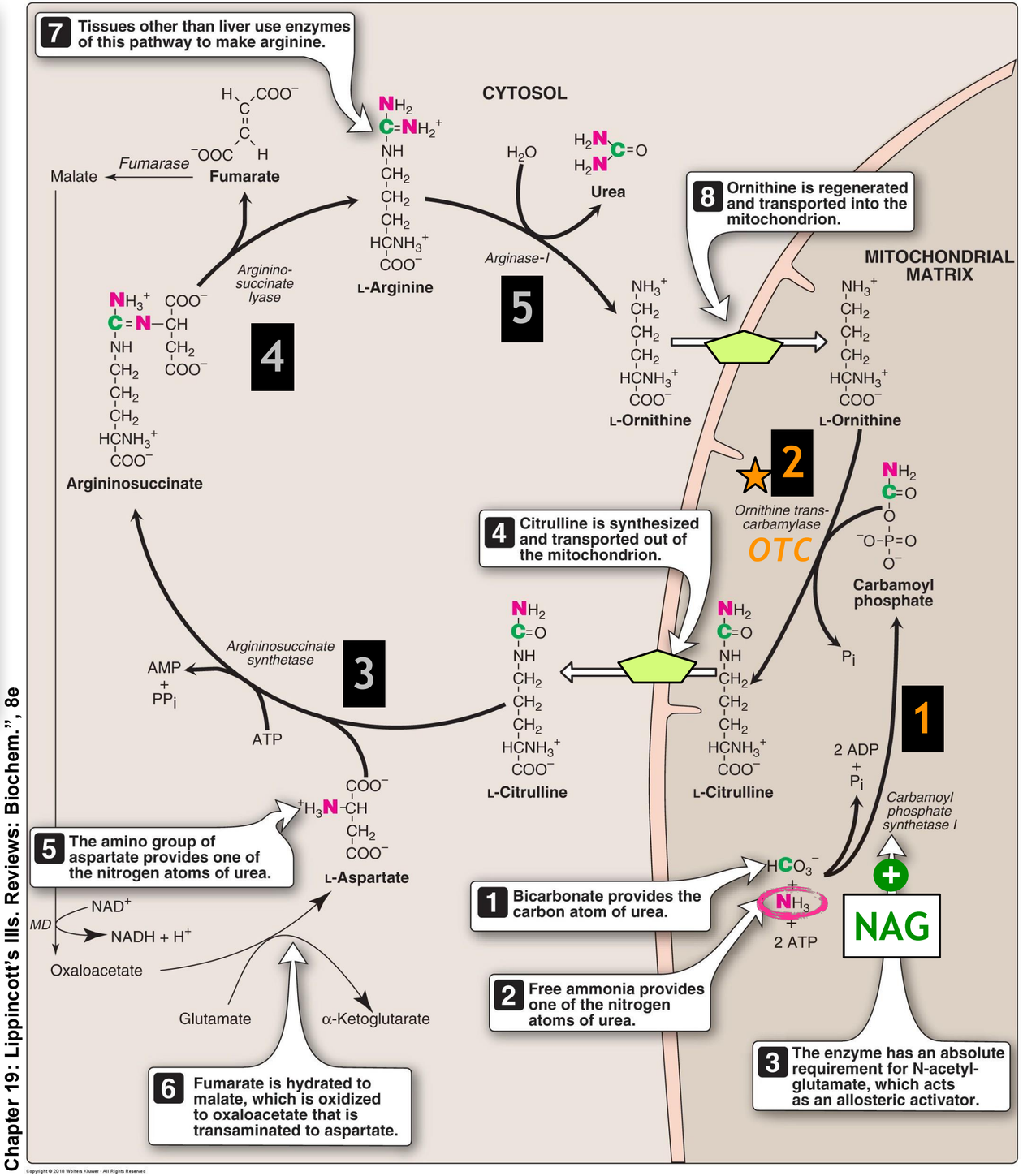


The ornithine transporter (ORNT1) moves ornithine across the inner mitochondria membrane.

OTC an X-linked recessive disorder, the other inborn errors of urea cycle are all autosomal recessive mutations.

Modified, BBS, 2024

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e

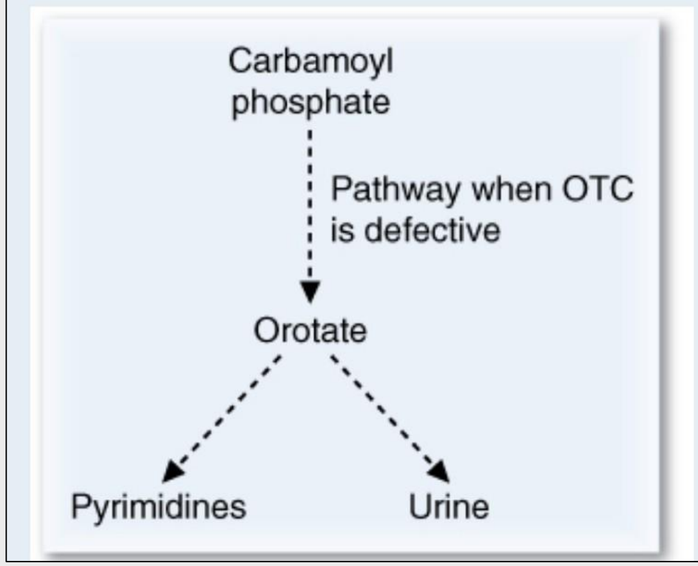
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Clinical correlation

★2 ***OTC deficiency is the most common inborn Urea cycle disorder:***

- ***X-linked***
- ***Frequency -1:80,000***

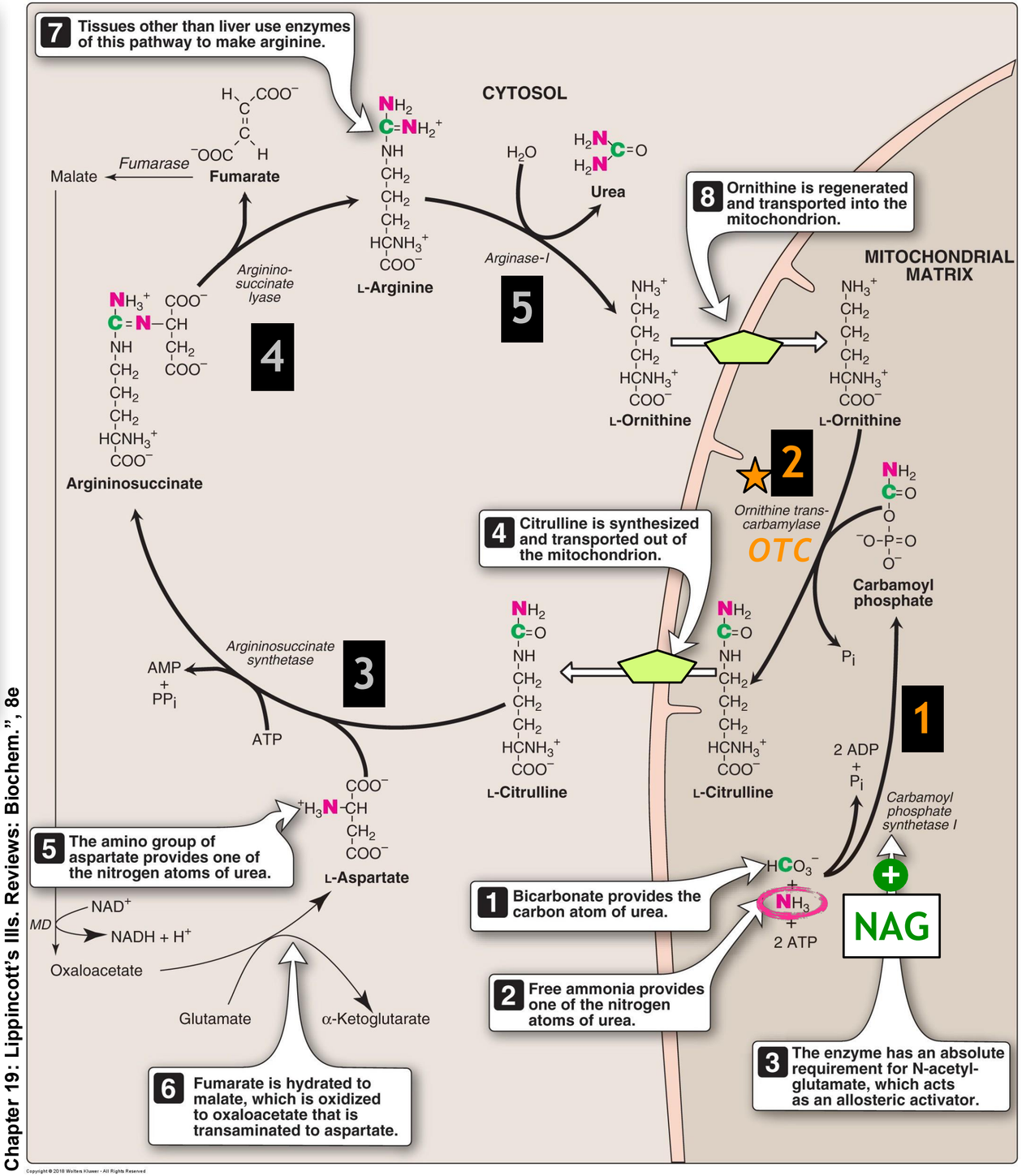
When OTC is deficient, the carbamoyl phosphate that normally would enter the urea cycle accumulates and floods the pathway for pyrimidine biosynthesis. Carbamoyl phosphate, produced by a cytoplasmic enzyme (CPSII), is the precursor for pyrimidine synthesis. Under these conditions, excess orotic acid (orotate), an intermediate in pyrimidine biosynthesis, is excreted in the urine. It produces no ill effects but is indicative of a problem in the urea cycle.



Lab Findings in OTC vs CPS1

Parameter	CPSI Deficiency	OTC Deficiency	Explanation
Ammonia	Very high	Very high	Both cause severe hyperammonemia due to urea cycle disruption
Orotic Acid	Normal	Elevated	OTC defect causes carbamoyl phosphate buildup, leading to increased orotic acid excretion
Glutamine	Elevated	Elevated	Elevated due to ammonia detoxification via glutamine synthesis
BUN	Low	Low	Urea cycle block reduces urea production, resulting in low BUN
Citrulline	Low	Low	Citrulline production is decreased due to block early in the urea cycle

The Urea Cycle



Modified from: Figures 19.14 and 19.16, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e

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Clinical Correlation

Treatments for Urea Cycle Defects

- ✓ Two priorities are:
  - Early diagnosis
  - Aggressive treatment to aid in nitrogen removal.
- ✓ In Argininosuccinate Lyase (ASL) deficiency, arginine supplementation has proved beneficial.
- ✓ This helps with regenerating ornithine and continue the cycle.
- ✓ The two nitrogen atoms are excreted as argininosuccinate

Hyperammonemia Causes

Hyperammonemia

Congenital

- ✓ **Caused by inherited defects in urea-cycle enzymes or associated proteins**
- ✓ **Impaired urea synthesis leads to hyperammonemia**, often presenting shortly after birth
- ✓ **Severe cases can cause irreversible** neurological damage and death if untreated

Acquired

- ✓ **Caused by significant liver dysfunction** – viral hepatitis, hepatotoxin, alcoholism, cirrhosis etc.
- ✓ **Impaired hepatic ammonia detoxification** leads to elevated blood ammonia levels
- ✓ **Elevated blood ammonia is** neurotoxic and can contribute to hepatic encephalopathy

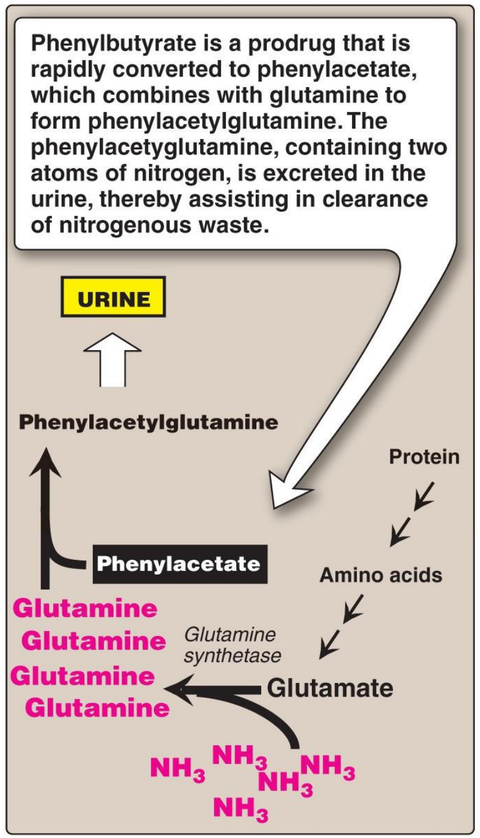
MUST KNOW

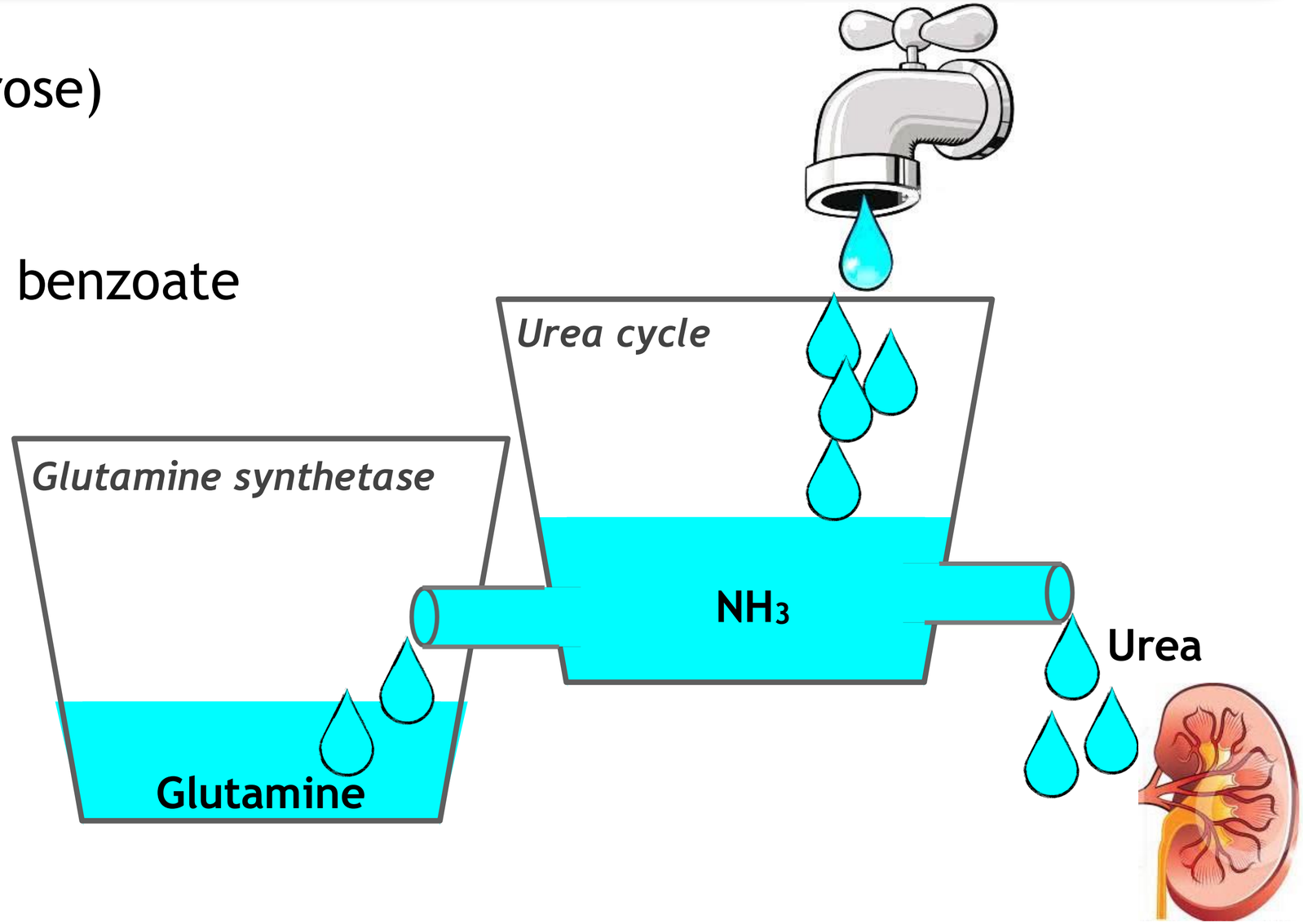
Mechanism of Neurotoxicity

- Excess **ammonia** promotes **glutamine synthesis** in **astrocytes** through the enzyme **glutamine synthetase**, leading to **intracellular glutamine accumulation**.
- The increased **glutamine concentration** contributes to **osmotic stress**, and **astrocyte swelling**. This swelling can contribute to **cerebral edema**, impaired neuronal function, and **neurological damage**.

Management of Hyperammonemia

- Low-protein diet
- High dose glucose (IV Dextrose)
- Hemodialysis
- Liver transplant
- Phenylbutyrate and sodium benzoate





Modified from: Figure 19.20, Chapter 19: Lippincott's Ills. Reviews: Biochem.", 8e